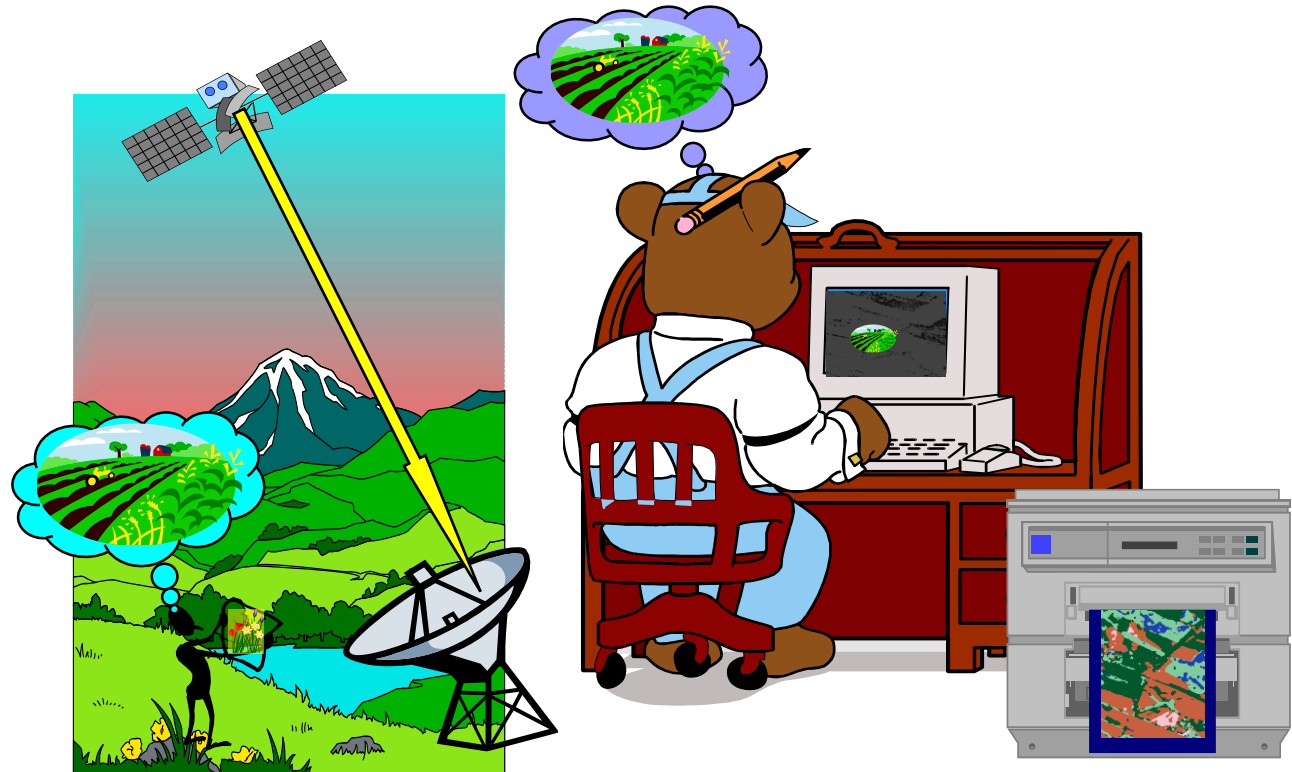


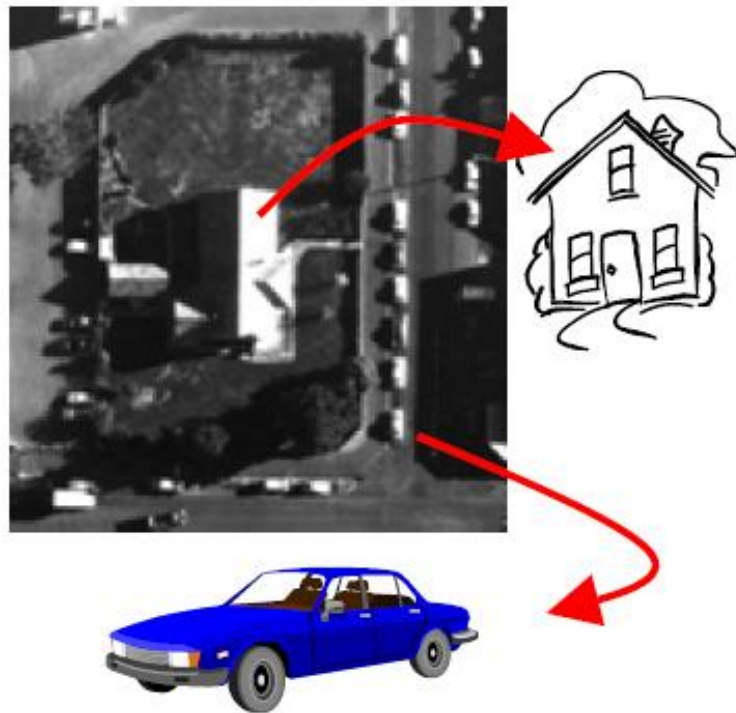
Concepts of Digital Image Analysis, Segmentation and OBIA



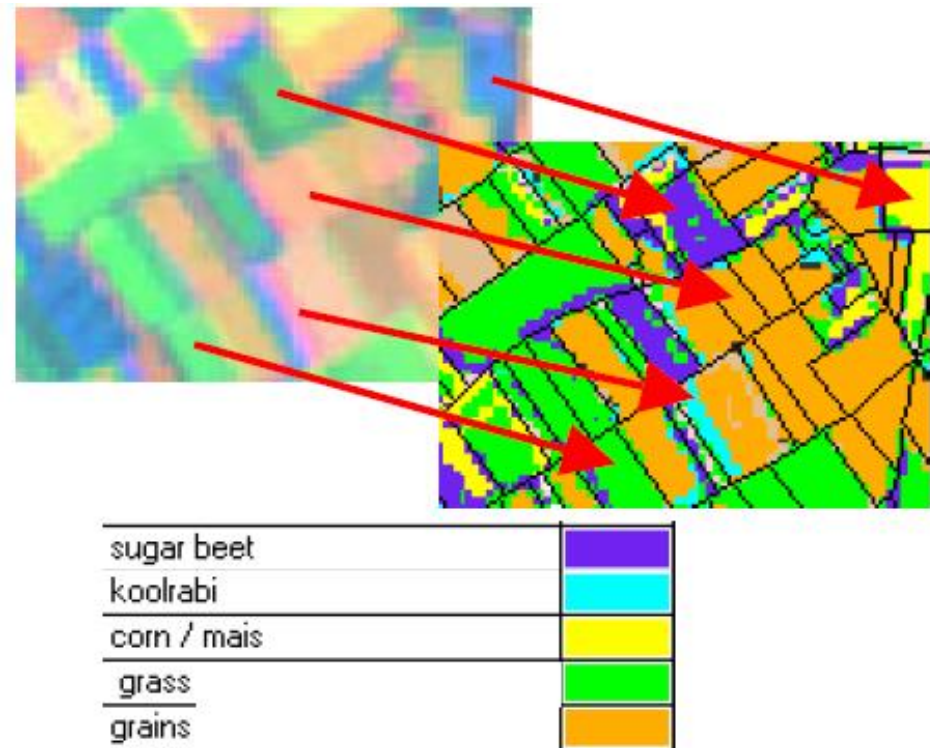
Dr. Poonam S. Tiwari

poonam@iirs.gov.in

Methods of Interpretation



Visual Interpretation



Digital Classification

Image Interpretation

- Image interpretation is defined as the act of examining images to identify objects and judge their significance

Elements of Image Interpretation

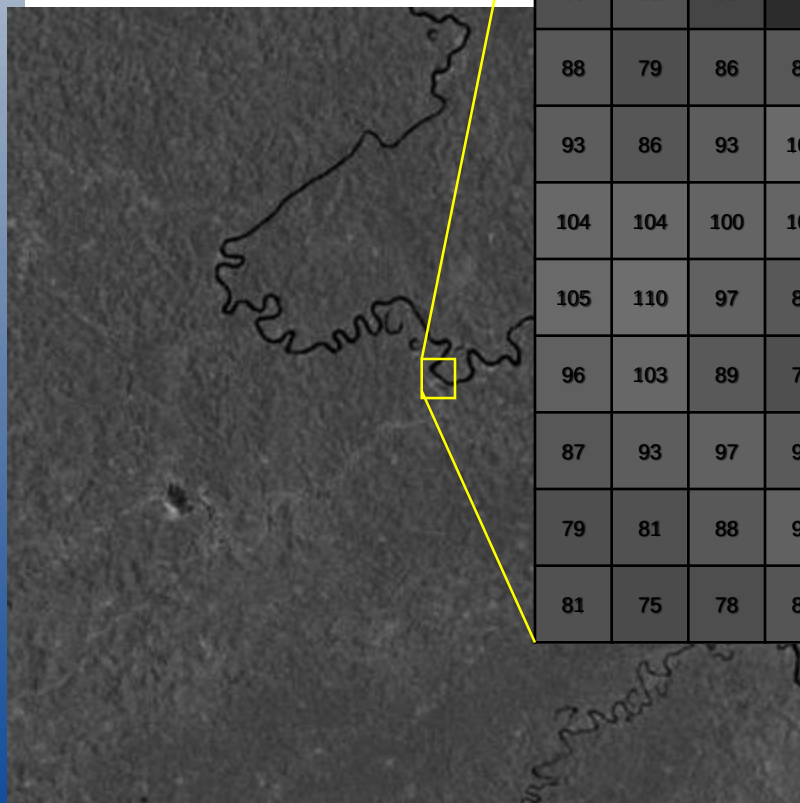
- Black and White Tone
- Color
- Size
- Shape
- Texture
- Pattern
- Height
- Shadow
- Association



What is a digital image?

- Grid cells or pixels
- Each pixel has a digital number (DN) which represents:

Spectral Reflectance Value



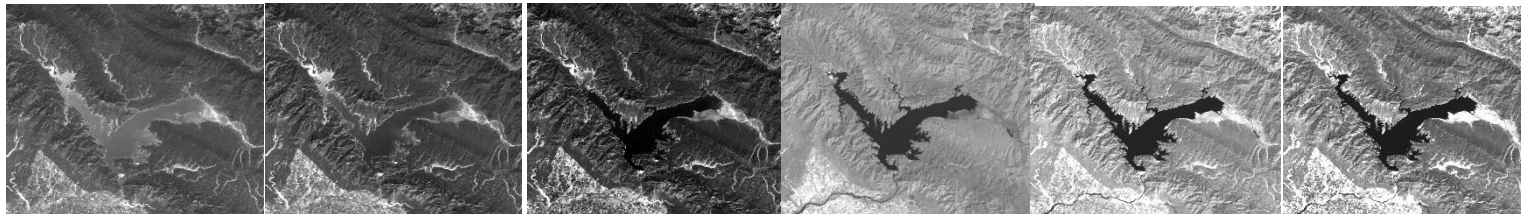
70	53	41	64	84	85	81	88	91	87
79	77	45	38	59	77	84	86	85	85
80	82	69	44	32	45	72	86	82	78
88	79	86	87	65	40	41	75	79	78
93	86	93	106	106	84	56	43	58	75
104	104	100	101	95	91	83	51	39	56
105	110	97	88	84	85	87	77	59	44
96	103	89	79	79	75	77	79	74	72
87	93	97	90	82	76	70	67	61	71
79	81	88	97	93	85	78	74	70	72
81	75	78	85	94	97	92	84	80	72

Pixel

Digital Number
(DN)

What your computer sees...

(0.45 to 0.515 μ m) (0.525 to 0.605 μ m) (0.63 to 0.690 μ m) (0.75 to 0.90 μ m) (1.55 to 1.75 μ m) (2.09 to 2.35 μ m)



Visible

Infrared

1

2

3

4

5

7

3,2,1

True Color Composite



(0.45 to 0.515 μ m) (0.525 to 0.605 μ m) (0.63 to 0.690 μ m) (0.75 to 0.90 μ m) (1.55 to 1.75 μ m) (2.09 to 2.35 μ m)



Visible

Infrared

1

2

3

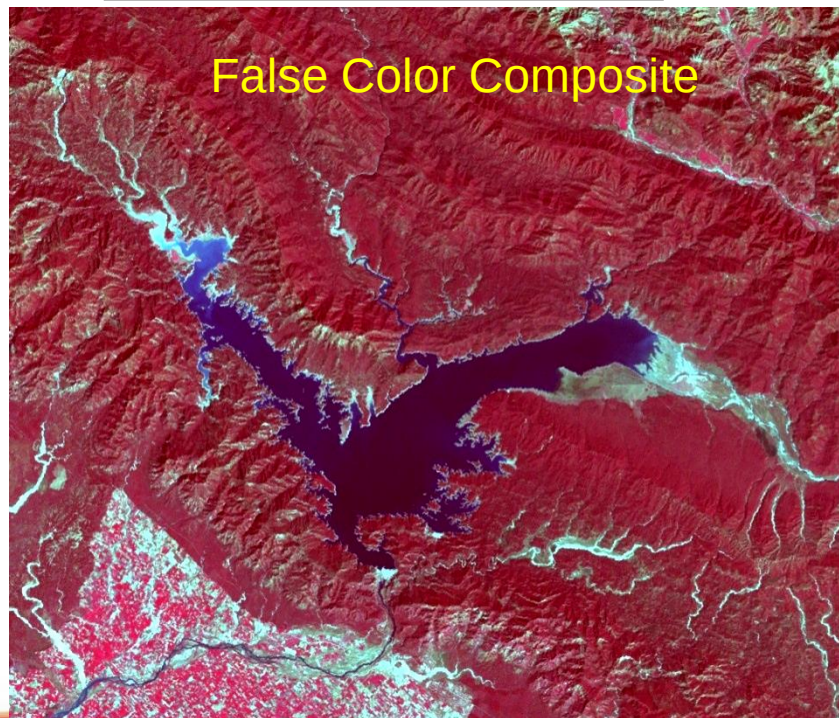
4

5

7

4,3,2

False Color Composite



Digital Image Processing

Digital image processing can be defined as *the computer manipulation of digital values contained in an image for the purposes of image correction, image enhancement and feature extraction.*

The process of DIP consist of mainly three steps :

☐ Data Acquisition/Restoration

Compensates for data errors, noise and geometric distortions introduced in the images during acquisitioning and recording

☐ Image Enhancement

Alters the visual impact of the image on the interpreter to improve the information content

☐ Information Extraction

Utilizes the decision making capability of computers to recognize and classify pixels on the basis of their signatures

☐ Others

Data merging, Change Detection, Hyperspectral image analysis...

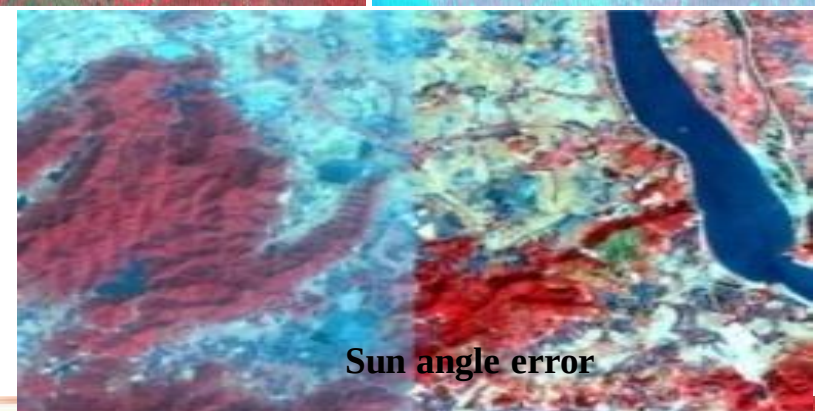
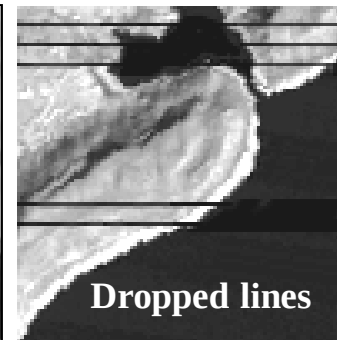
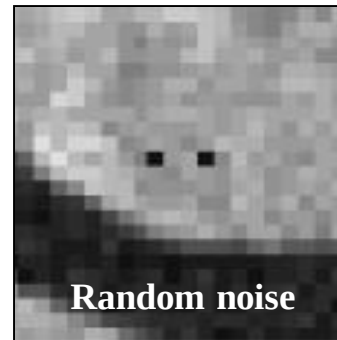
Image Preprocessing

- *Remote sensing systems do not function perfectly all the time .*
- Pre-processing operations, referred to as image restoration and rectification, are intended to correct for sensor- and platform-specific radiometric and geometric distortions of data.
- Pre processing functions are normally carried out prior to the main data analysis and extraction of information
- The two most common types of error encountered in remotely sensed data are
 - *radiometric* and
 - *geometric*.

Radiometric errors and Correction

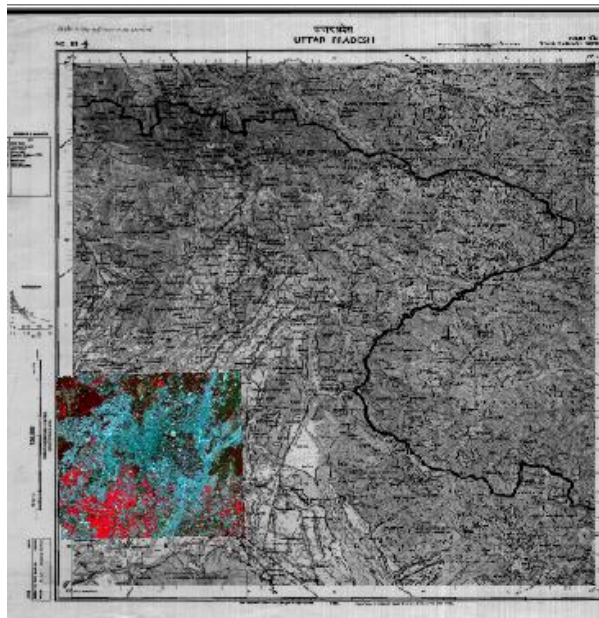
Several radiometric errors are:

- ❑ Periodic line or column drop-outs,
 - ❑ Correction – Replacement by preceding, succeeding or average
- ❑ line or column striping.
 - ❑ Correction – Sensor Model, Histogram Matching
- ❑ random bad pixels (shot noise),
 - ❑ Correction – Filtering
- ❑ partial line or column drop-outs
- ❑ Atmospheric Errors – Haze
 - ❑ Correction – Dark Object Subtraction



Geometric Error Correction

- The transformation of remotely sensed images so that it has a scale and projections of a map is called geometric correction.
- It is necessary when accurate area , distance and direction measurements are required to be made from the imagery.
- It is achieved by transforming the data from one grid system into another grid system using a geometric transformation

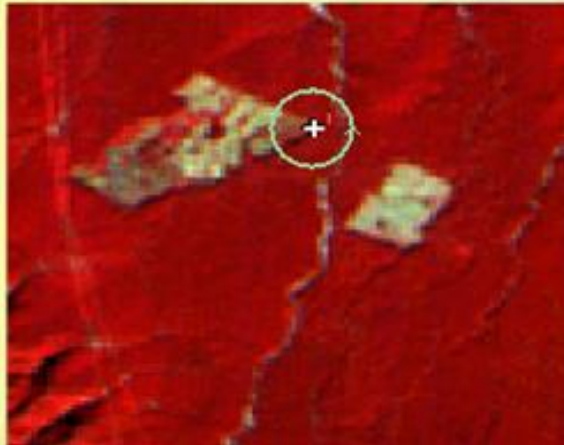


- **Spatial Interpolation**
- **Intensity Interpolation**
 - Nearest neighbor
 - Bilinear interpolation
 - Cubic Convolution

Ground Control Points (GCPs)

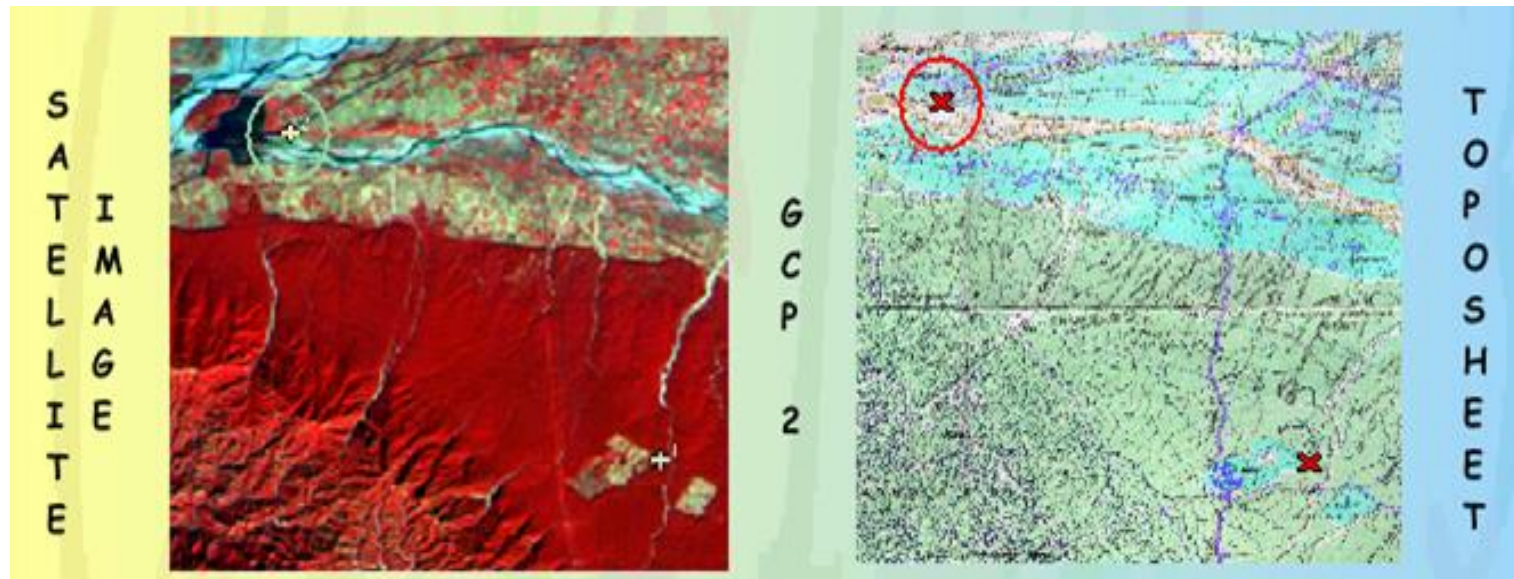
- ❑ Road intersections, river bends, distinct natural features, etc.
- ❑ GCPs should be spread across image
- ❑ Requires a minimum number depending on the type of transformation

First some GCP's (Ground Control Points) are selected. These are certain reference points, that are bright enough and can be easily located. We take the satellite image as well as a reference data of the same area that can be either a toposheet or a reference image using GPS. Shown below is an example of that.....

S
A
T
E
L
L
I
T
E

G
C
P

1

T
O
P
O
S
H
E
E
T



For solving the first order transformation equations, we need a minimum of three GCP's. For more accuracy and root mean square adjustment, a fourth GCP is also selected. We can even choose more GCP's according to the requirements.

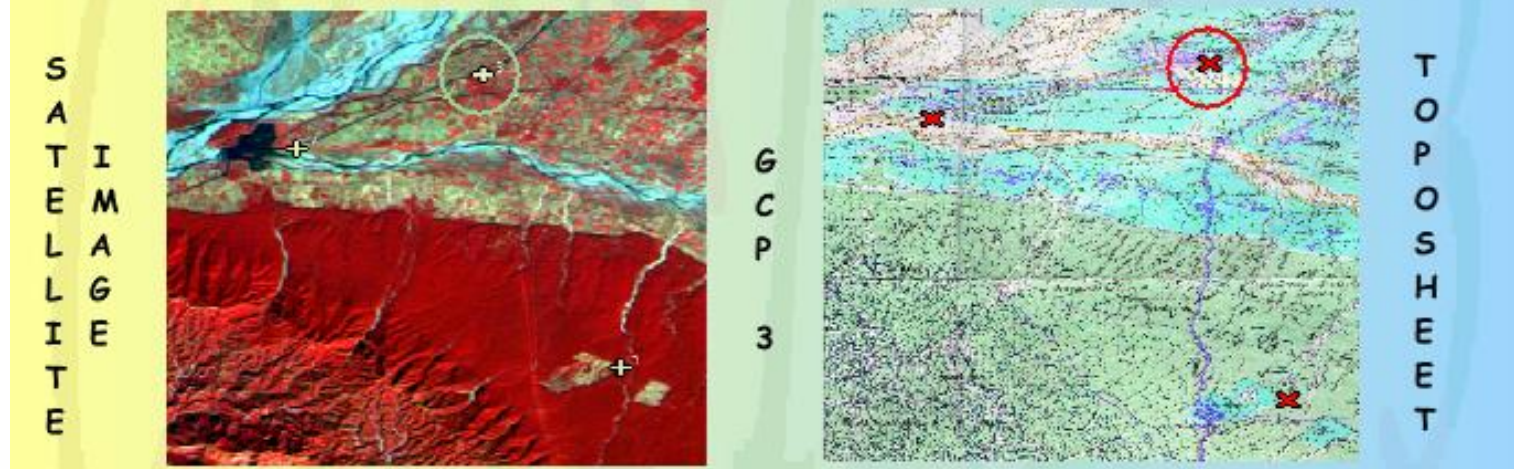


Image Rectification

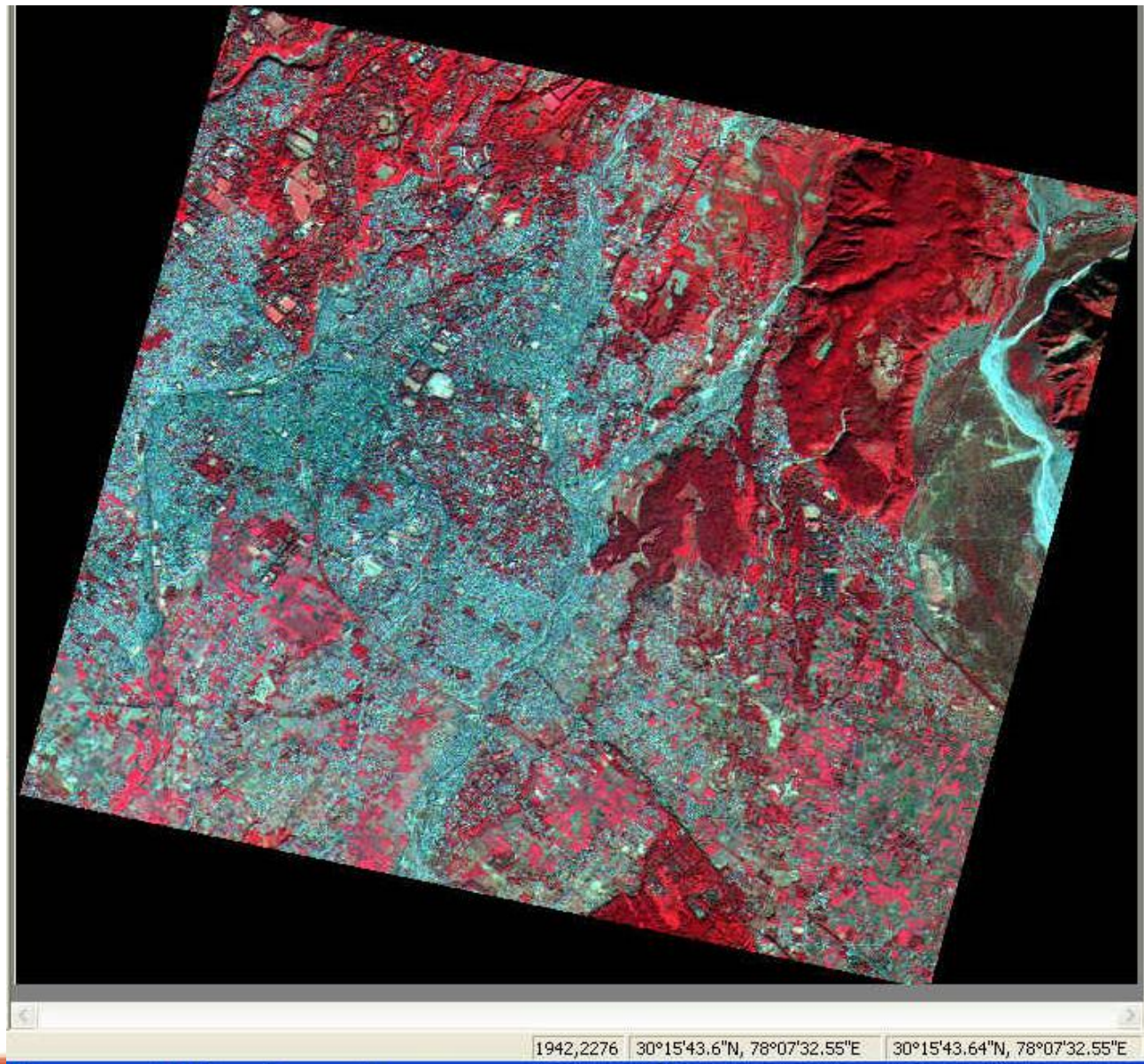
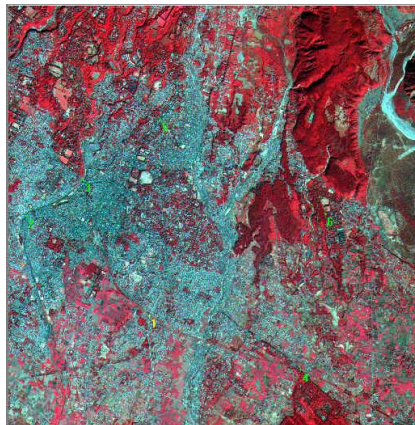
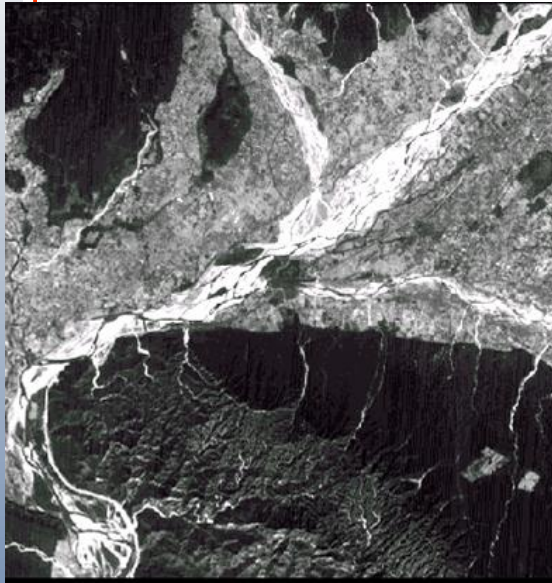


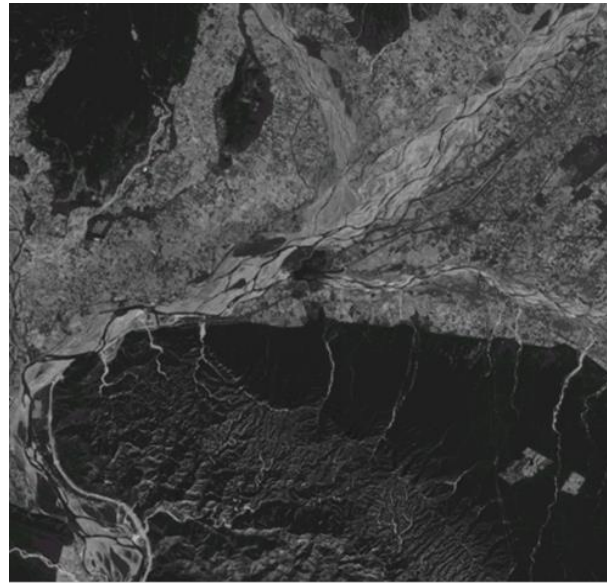
Image Enhancement

- Modification of an image to alter its impact on viewer
 - Enhancements are used to make it easier for visual interpretation and understanding of imagery.
- Process of making an image more interpretable for a particular application
 - to accentuate certain image features for subsequent analysis or for image display

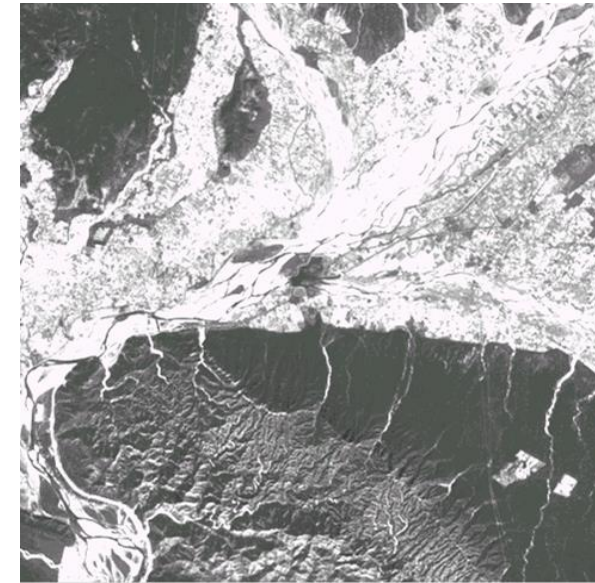
Image and Contrast



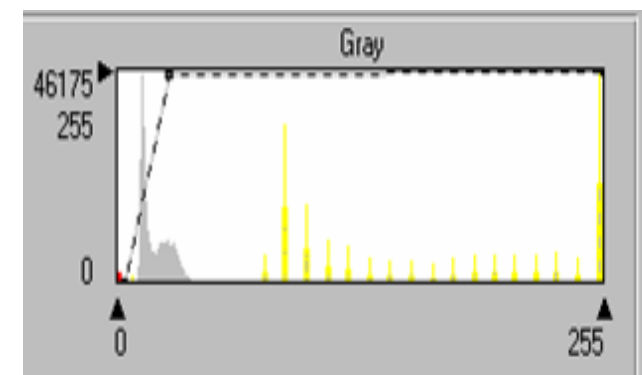
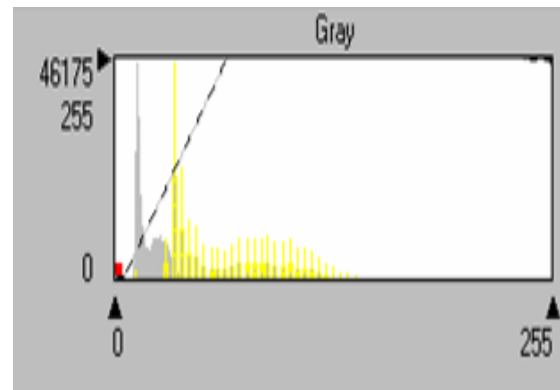
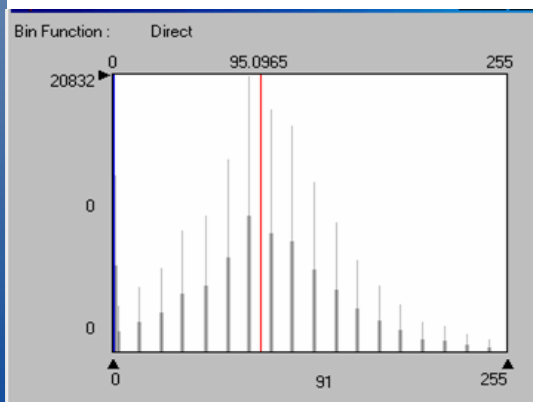
Original



Low Contrast

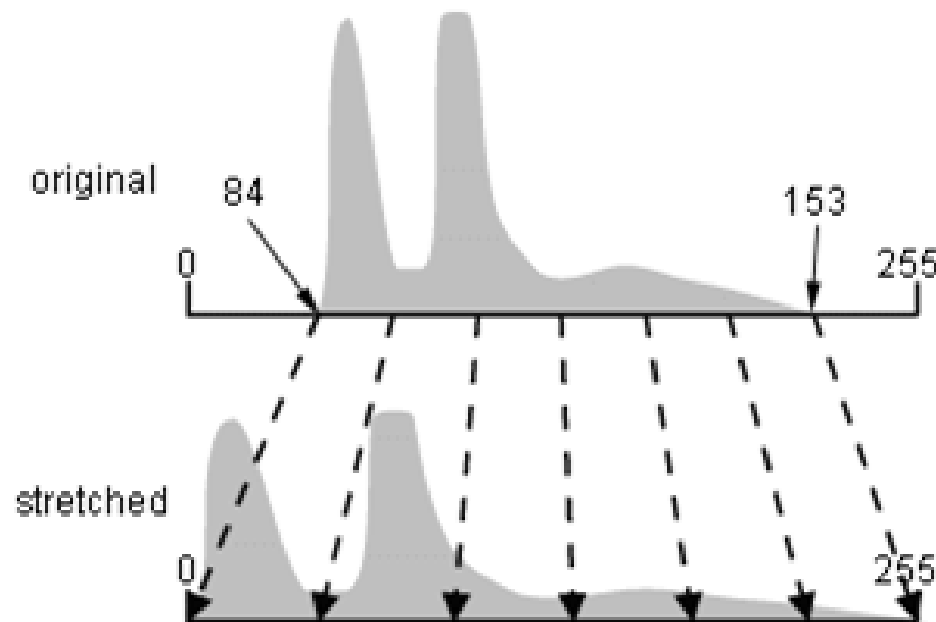


Saturated



Linear Contrast Stretch

- A DN in the low range of the original histogram is assigned to extreme black, and a value at the high end is assigned to extreme white.
- The remaining pixel values are distributed linearly between these two extremes



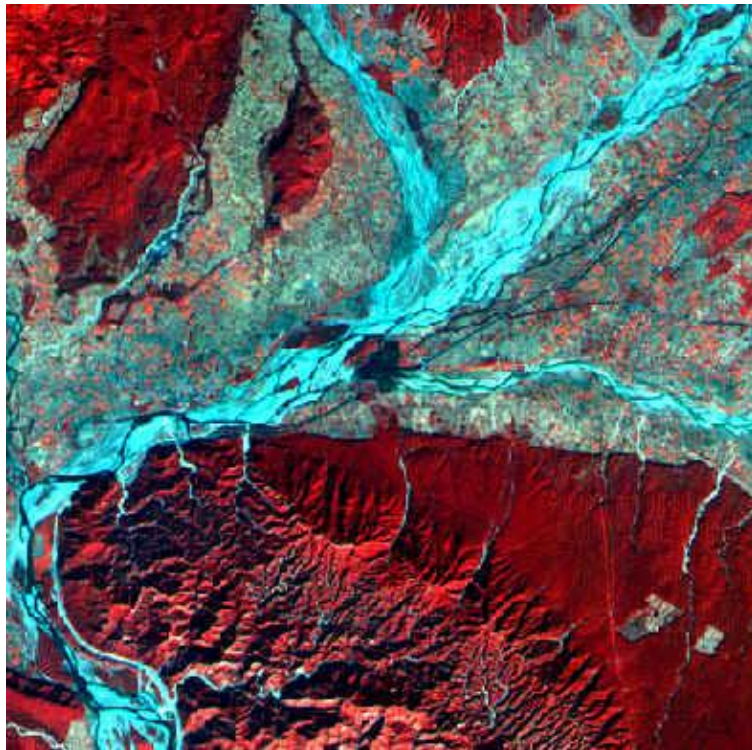
Filtering

- pixel value is modified based on the values surrounding it.
- Process of suppressing (de-emphasizing) certain frequencies & passing (emphasizing) others.
- This technique increases the analyst's ability to discriminate detail.
- used for enhancing certain features , for removal of noise and Smoothening of image

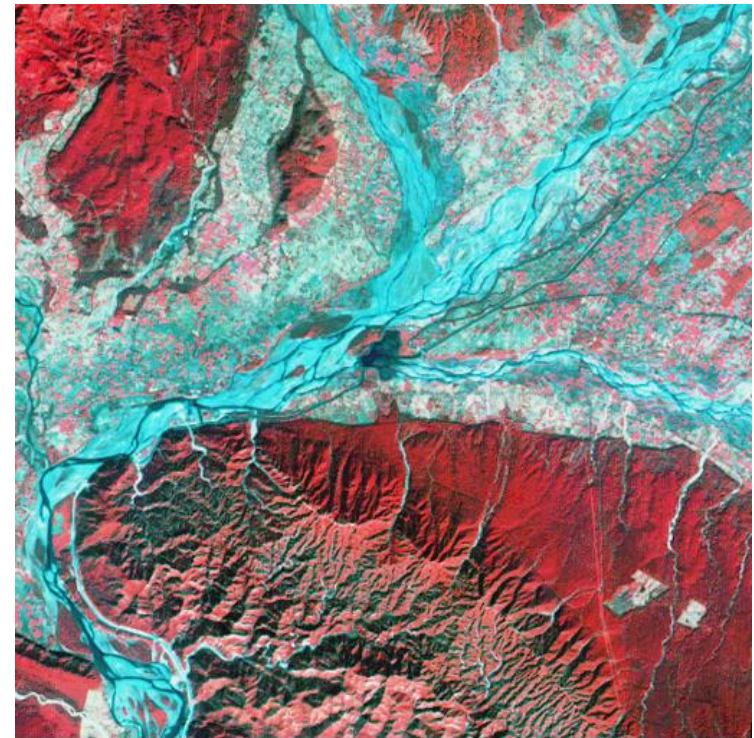
Filter Types	
Low Pass Filters	High Pass Filters
block high frequency details	Preserves high frequencies and Removes slowly varying components
has a smoothening effect on images	Emphasizes fine details
Used for removal of noise Removal of “salt & pepper” noise	Used for edge detection and enhancement
Blurring of image especially at edges	Edges - Locations where transition from one category to other occurs

Low Pass Filtering

A **low-pass filter** is designed to emphasize larger, homogeneous areas of similar tone and reduce the smaller detail in an image. Thus, low-pass filters generally serve to smooth the appearance of an image. Average and median filters, often used for radar imagery, are examples of low-pass filters.



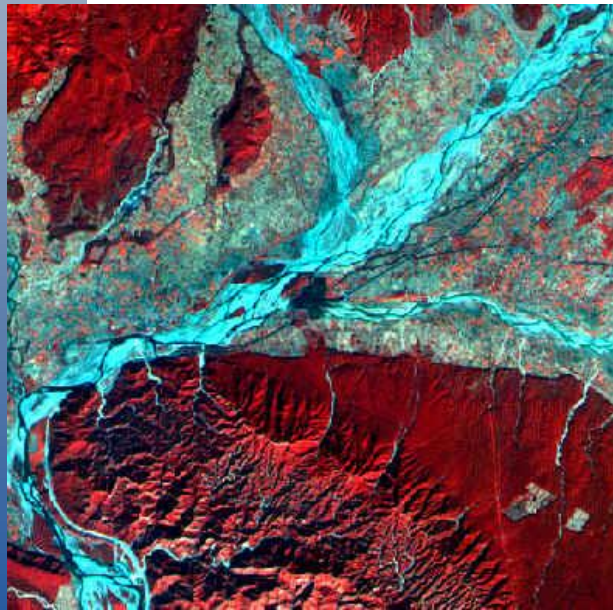
FCC



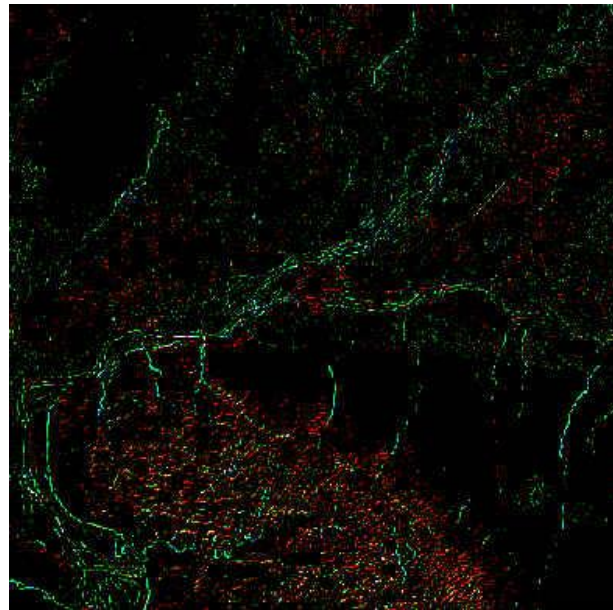
Low Pass Filtered Image

High Pass Filtering

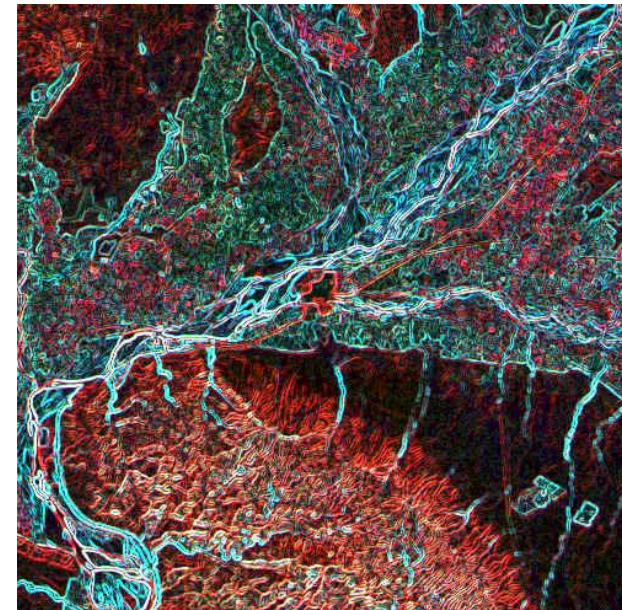
High-pass filters do the opposite and serve to sharpen the appearance of fine detail in an image. One implementation of a high-pass filter first applies a low-pass filter to an image and then subtracts the result from the original, leaving behind only the high spatial frequency information



FCC



Edge Detector



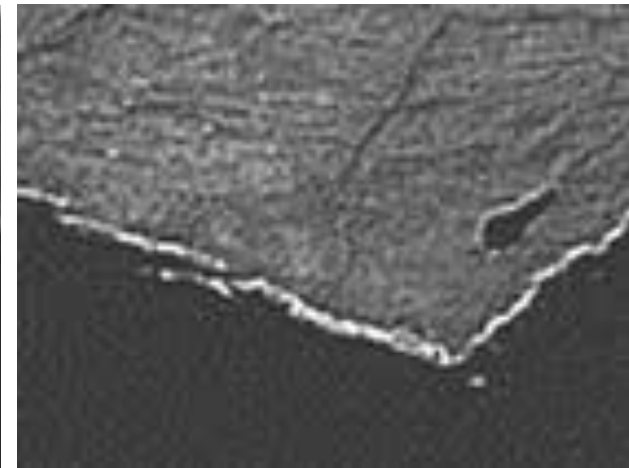
Edge Enhance

Directional High Pass Filtering

Directional, or edge detection filters are designed to highlight linear features, such as roads or field boundaries.

These filters can also be designed to enhance features which are oriented in specific directions.

These filters are useful in applications such as geology, for the detection of linear geologic structures.



Multi Band Operations

Vegetation Indices

These are computed to eliminate the changing illumination conditions, surface slope, aspect, and other extraneous factors. The vegetation indices are helpful in highlighting the vegetation content in the imagery.

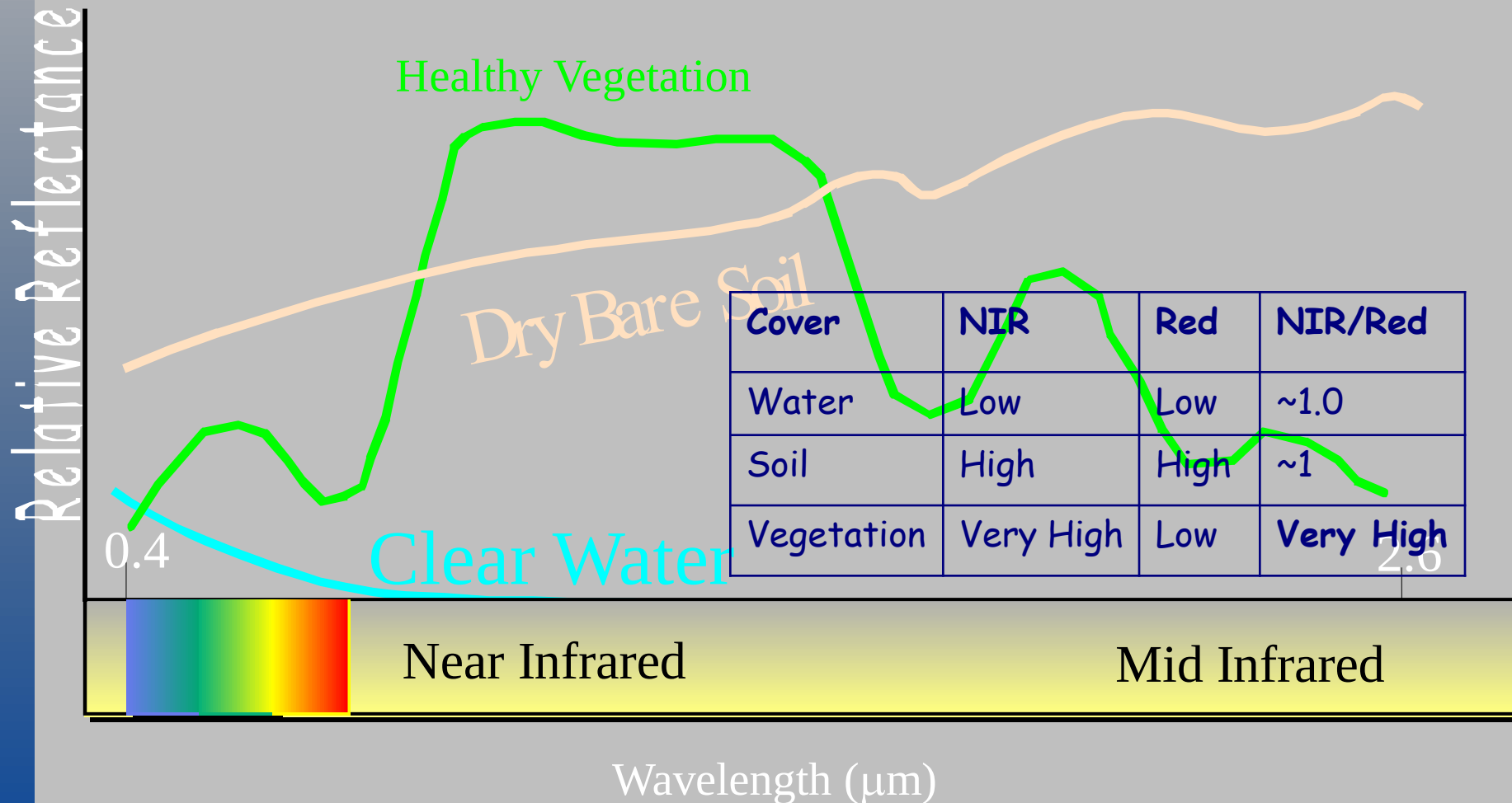
The commonly used vegetation indices are:

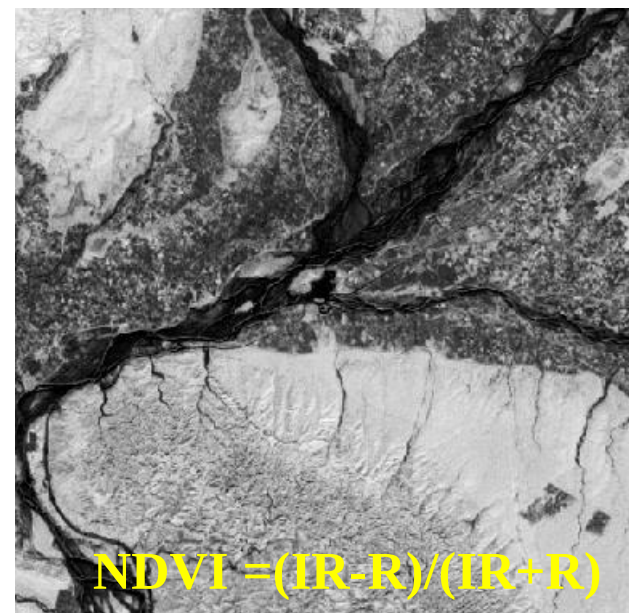
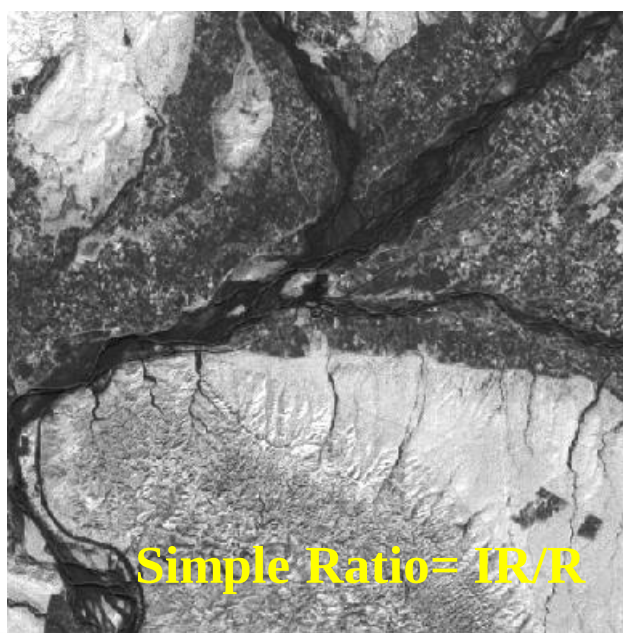
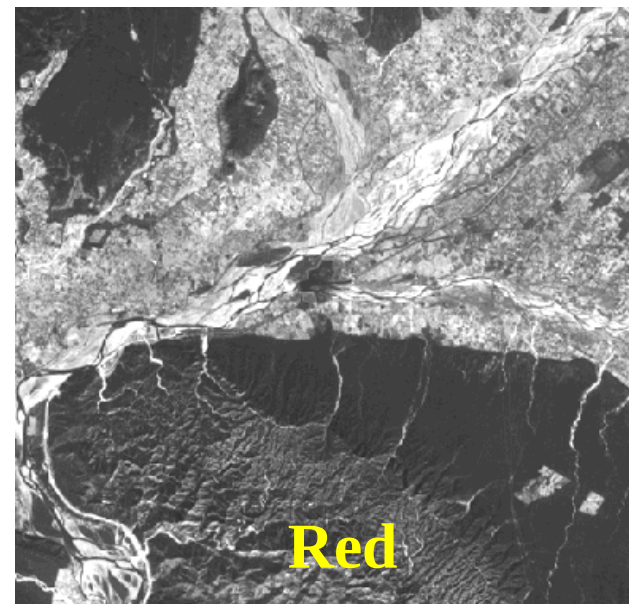
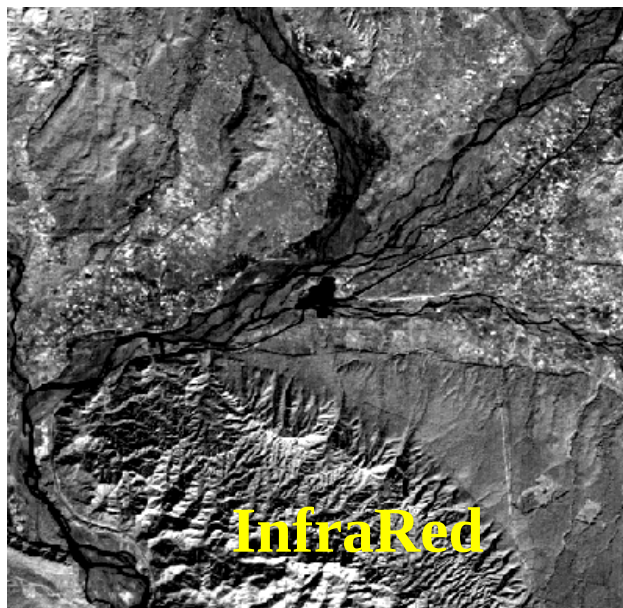
$$\text{Vegetation Index (VI)} = \text{IR} / \text{R}$$

$$\begin{aligned} \text{Normalized Difference Vegetation Index (NDVI)} \\ = (\text{IR} - \text{R}) / (\text{IR} + \text{R}) \end{aligned}$$

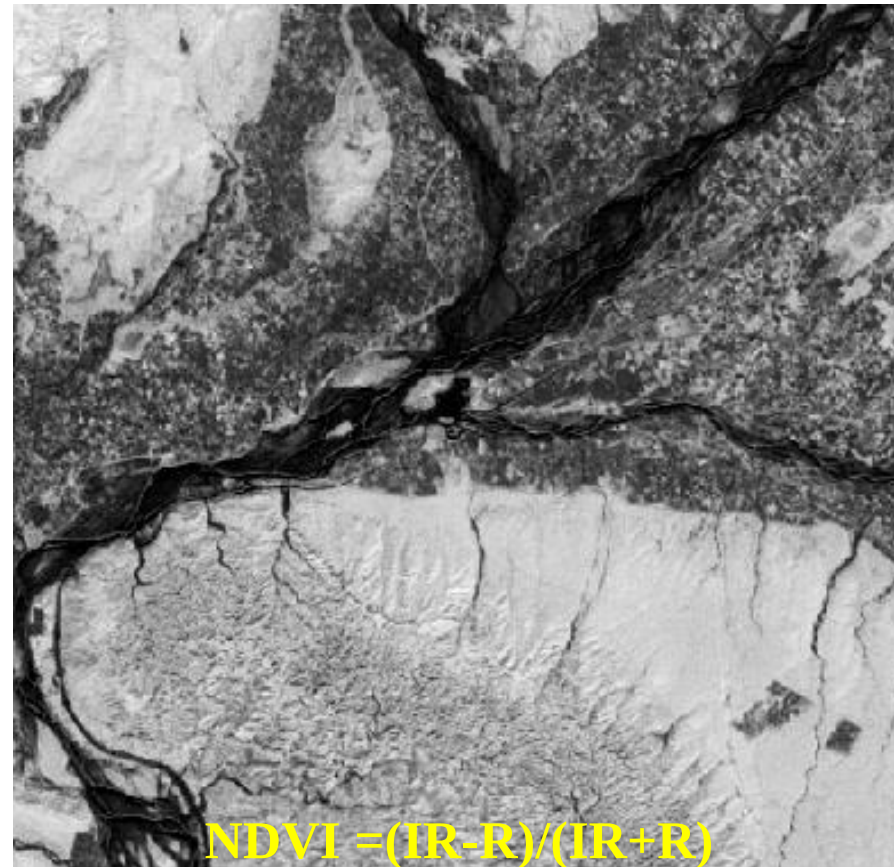
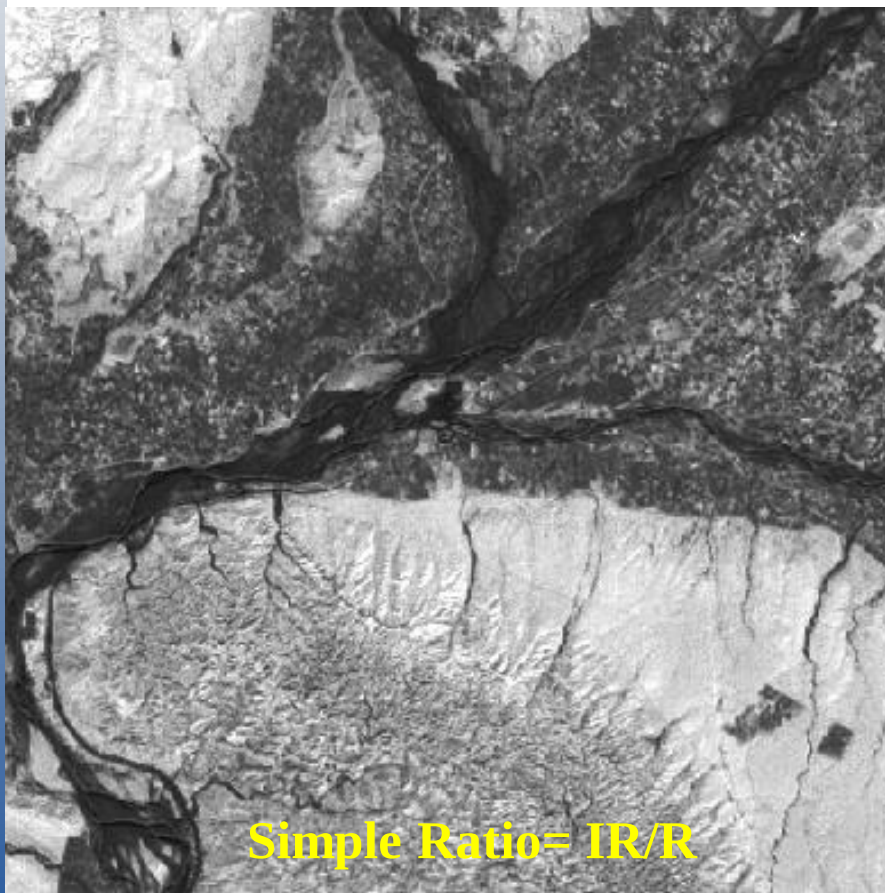
Why Ratio

The most unique feature of the vegetation reflectance curve is the low red reflectance and high NIR reflectance.





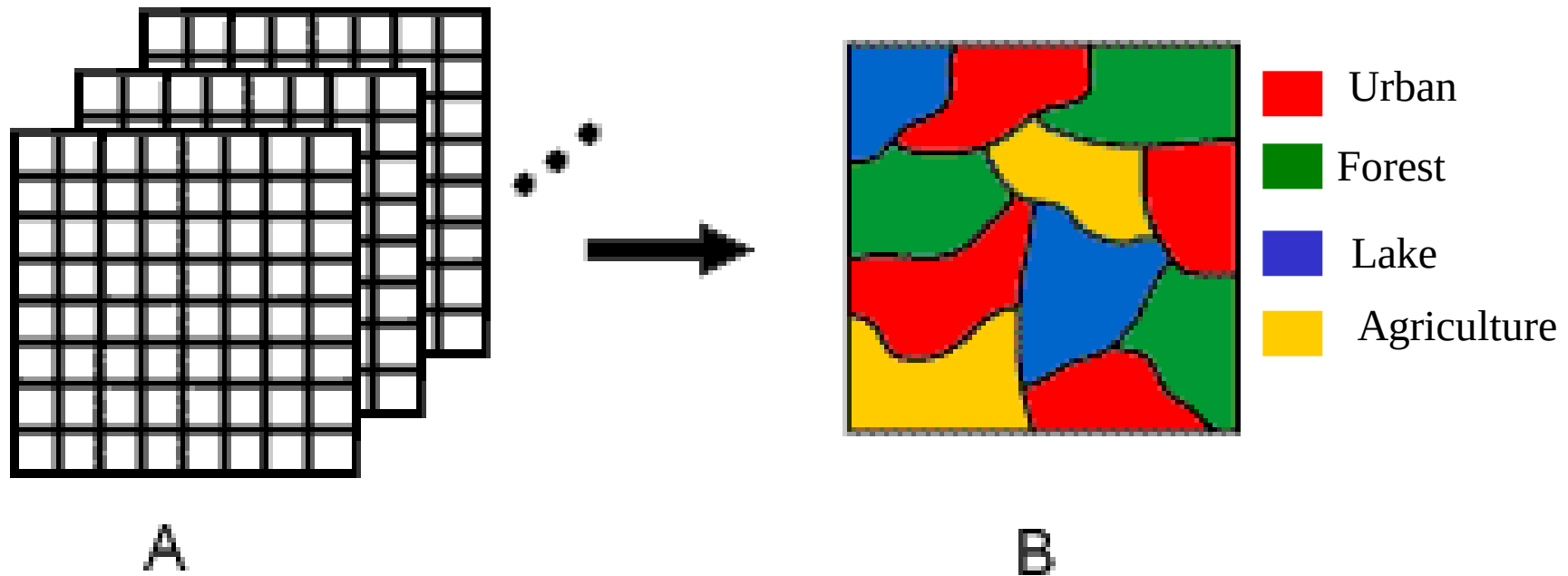
Vegetation Index



Multispectral Classification

- Assigning each pixel in a remotely sensed image a label describing real world object.
- That is automatically categorizing all pixels in an image into land cover classes or themes.
- uses the spectral information represented by the digital numbers in one or more spectral bands, and attempts to classify each individual pixel based on this spectral information.
- Output is a classified map- a form of digital thematic map

Multispectral Classification



Assigning each pixel in a remotely sensed image a label describing real world object

Classification Types

Common classification procedures can be broken down into two broad subdivisions based on the method used

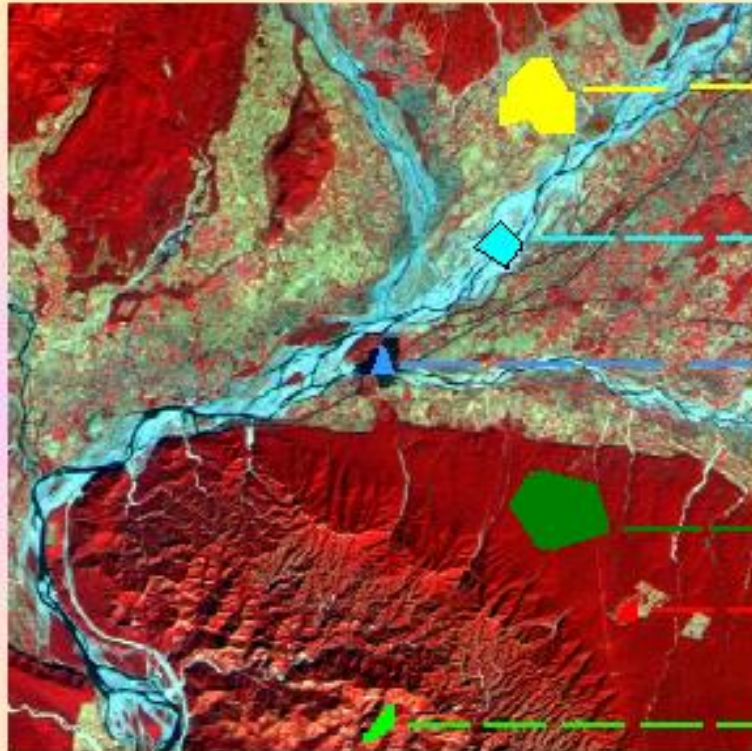
- Supervised classification
- Unsupervised classification

Supervised Classification

- Samples of known identity are used to classify pixels of unknown identity
- Multivariate statistical parameters are calculated for the known samples.
- Every pixel is evaluated and assigned to the class which it most closely resemble digitally (in statistics).
- Hard Classification – a pixel is assigned to only one class.
- Called as supervised as the analyst supervises the algorithm by providing training sites

Supervised Classification

UNCLASSIFIED IMAGE



The statistically calculated data of the matrices:

$$\begin{array}{lll} \mu_{11} = 35 & \mu_{21} = 30.22 & \mu_{31} = 38.11 \\ s = 1.25 & s = 1.56 & s = .78 \end{array}$$

$$\begin{array}{lll} \mu_{12} = 35.44 & \mu_{22} = 38.89 & \mu_{32} = 50.11 \\ s = 1.42 & s = 1.69 & s = 1.76 \end{array}$$

$$\begin{array}{lll} \mu_{13} = 30.89 & \mu_{23} = 11.89 & \mu_{33} = 10.11 \\ s = .33 & s = .33 & s = .33 \end{array}$$

$$\begin{array}{lll} \mu_{14} = 34.89 & \mu_{24} = 11.89 & \mu_{34} = 27.67 \\ s = .33 & s = .33 & s = .5 \end{array}$$

$$\begin{array}{lll} \mu_{15} = 31.55 & \mu_{25} = 21.56 & \mu_{35} = 33.56 \\ s = 1.5 & s = 1.33 & s = 1.42 \end{array}$$

$$\begin{array}{lll} \mu_{16} = 48.667 & \mu_{26} = 19.56 & \mu_{36} = 32.33 \\ s = 3 & s = 1.74 & s = 1 \end{array}$$

A – Agriculture

F – Forest

DR – Dry River

U – Urban Area

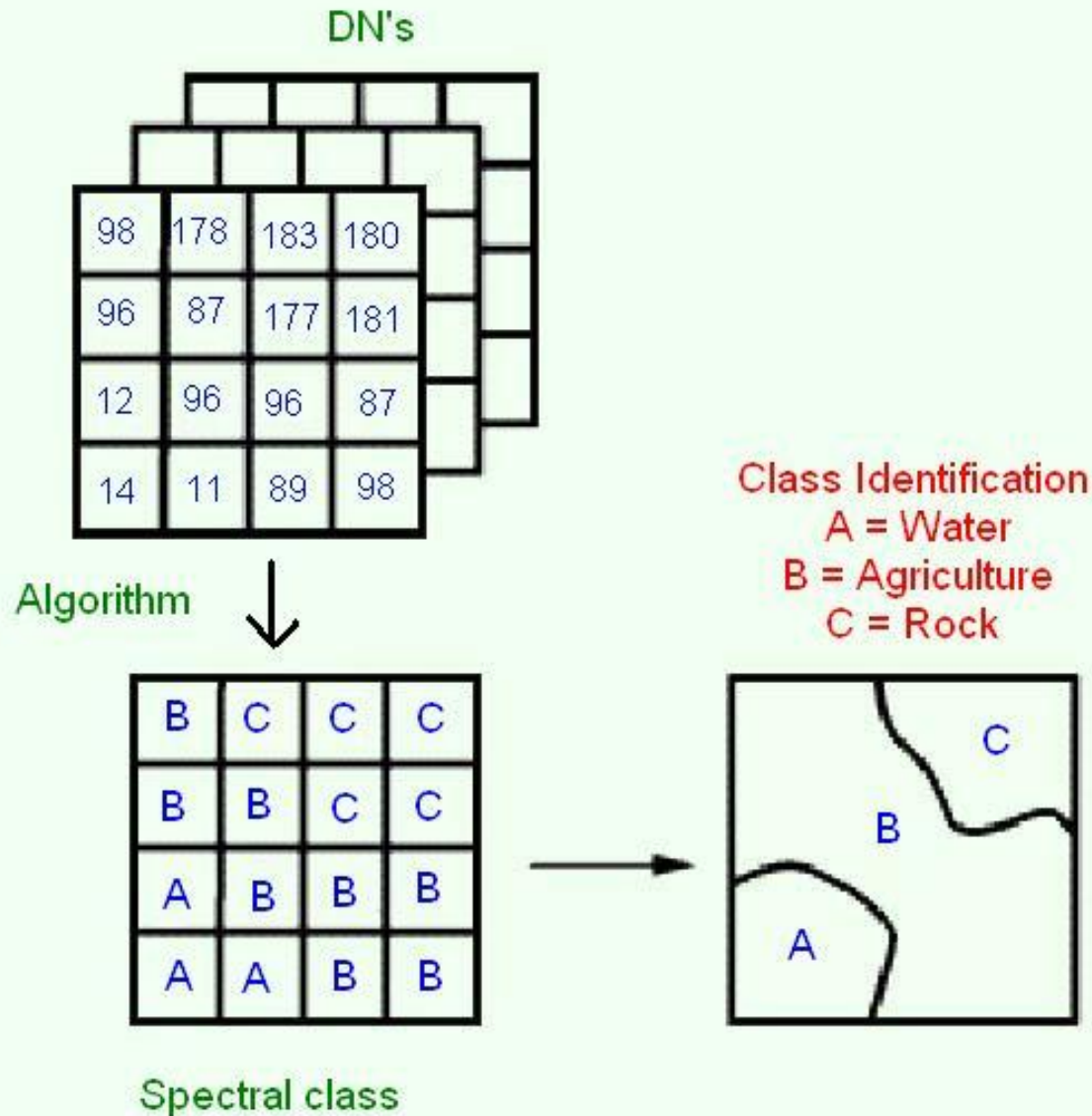
W – Reservoir

OF – Open Forest

Unsupervised Classification

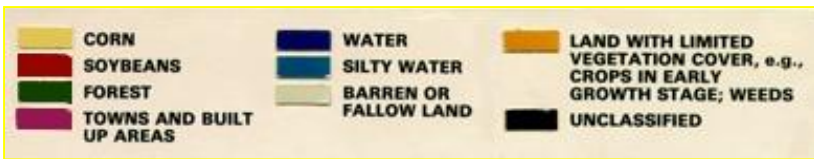
- The identities of land cover types to be specified as classes within a scene are generally not known a priori because ground reference information is lacking or surface features within the scene are not well defined.
- The computer is required to group pixels with similar spectral characteristics into unique clusters according to some statistically determined criteria.
- Analyst then combine and re-labels the spectral clusters into information classes.

Unsupervised Classification



Accuracy Assessment

- Comparison to two sources of information
 - Remote Sensing derived classification map
 - Reference Test information
- The relationship between the two sets of information's is expressed as a matrix known as
 - ERROR MATRIX / CONFUSION MATRIX / CONTINGENCY TABLE



From airphotos, field data, etc.

- 1 Generate a random sample of points
- 2 For each point, compare the map class with the “true” class
- 3 Construct and analyze an error matrix...

Accuracy Assessment: the error matrix

- Reference data labels: the class label of the “truth” sample point derived from data collected that is assumed to be correct (columns).
- Map labels: the category label, of the same sample point, derived from the satellite image classification (rows).

		REFERENCE DATA				Total possible
		Corn	Soybeans	Forest	Other	
CLASSIFIED IMAGE	Corn	25	5	10	3	43
	Soybeans	2	50	6	5	63
	Forest	3	4	60	5	72
	Other	2	2	2	100	106
	Total	32	61	78	113	284

- * Number of correctly classified pixels shown along the major diagonal
- * Off-diagonals represent errors of omission and commission

Accuracy Assessment

the error matrix

■ Representations of map accuracy:

- overall accuracy (of the classification or map)

$$\frac{25 + 50 + 60 + 100}{284} * 100 = 82.7\%$$

CLASSIFIED IMAGE

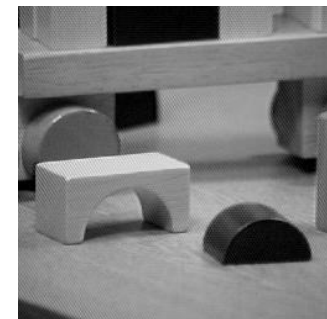
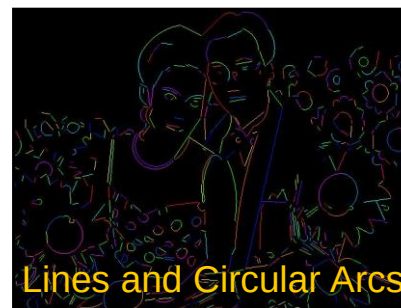
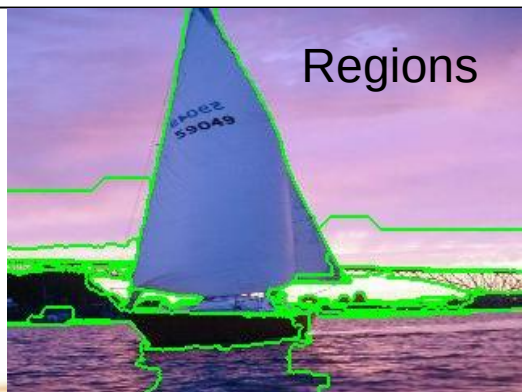
	REFERENCE DATA				Total Possible
	Corn	Soybeans	Forest	Other	
Corn	25	5	10	3	43
Soybeans	2	50	6	5	63
Forest	3	4	60	5	72
Other	2	2	2	100	106
Total	32	61	78	113	284

- * Number of correctly classified pixels (sum of the major diagonal)
- * Divided by the number of reference pixels

Image Segmentation

- The process of dividing an image into regions of related content.
- The process of partitioning an image into some non-overlapping meaningful homogeneous regions.
- Image segmentation is the operation of partitioning an image into a collection of connected sets of pixels

1. into regions, which usually cover the image
2. into linear structures, such as
 - line segments
 - curve segments
3. into 2D shapes, such as
 - circles
 - ellipses
 - ribbons (long, symmetric regions)



Need: High resolution satellite images have richer spatial but less spectral content.

Main Methods of Region Segmentation

1. Thresholding

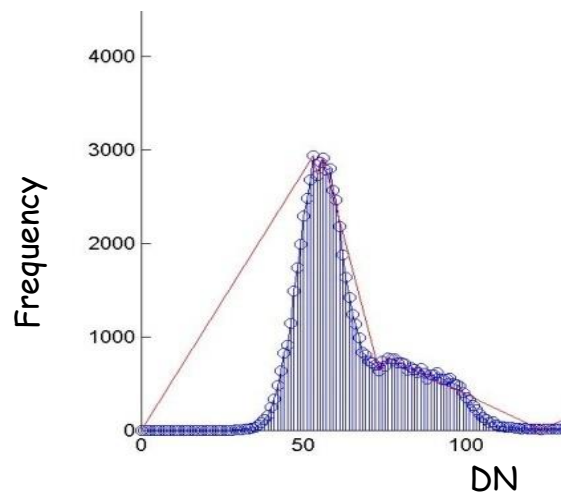
2. Region Growing

3. Clustering

4. Split and Merge

Histograms

- Histogram are constructed by splitting the range of the data into equal-sized bins (called classes). Then for each bin, the number of points from the data set that fall into each bin are counted.
- Vertical axis: Frequency (i.e., counts for each bin)
- Horizontal axis: Response variable
- In image histograms the pixels form the horizontal axis



THRESHOLDING TECHNIQUE

- Determining Histogram
- Choosing threshold
- Labeling or classification
- Extraction of blob by connecting components

Histogram-based threshold

- Compute the gray level histogram of the image.
- Find two “clusters”: black and white.
- Minimizing the L_2 error:
 - Select initial estimate T
 - Segment the image using T .
 - Compute the average gray level of each segment m_b, m_w
 - Compute a new threshold value: $T = \frac{1}{2} (m_b + m_w)$
 - Continue until convergence.

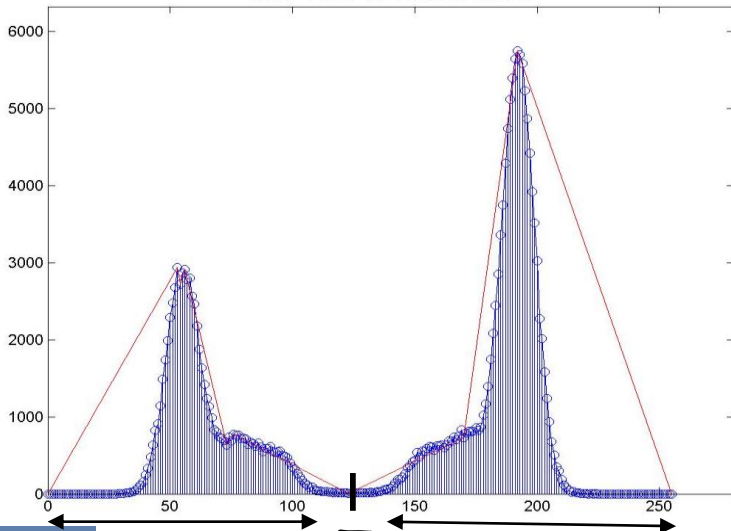
Thresholding

- Suppose that the gray-level histogram corresponds to an image, $f(x,y)$, composed of dark objects in a light background, in such a way that object and background pixels have gray levels grouped into two dominant modes.
- One obvious way to extract the objects from the background is to select a threshold 'T' that separates these modes.
- Then any point (x,y) for which $f(x,y) > T$ is called an object point, otherwise, the point is called a background point.

Bimodal Histogram

- If two dominant modes characterize the image histogram, it is called a bimodal histogram.
- Only one threshold is enough for partitioning the image.

Original Points: 256 Evolved Points: 10



Background or Object 1

Foreground or Object 2

Gray Scale Image - bimodal

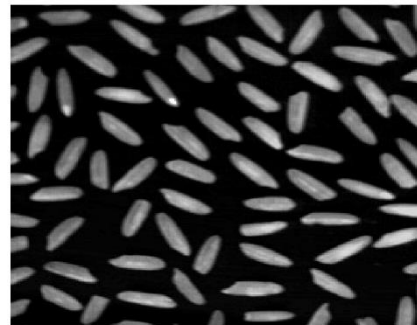


Image of rice with black background

Original Points: 256 Evolved Points: 10

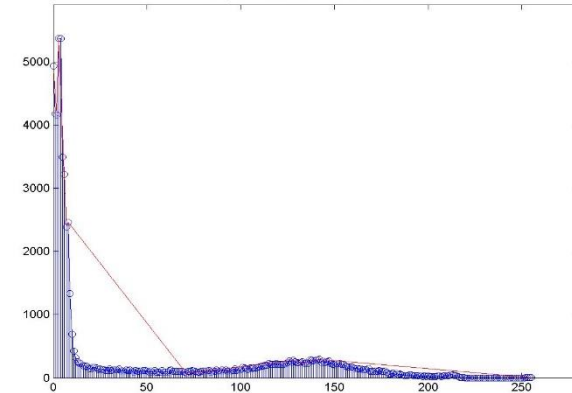


Image histogram of rice



Image of a Finger Print with light background

Original Points: 256 Evolved Points: 10

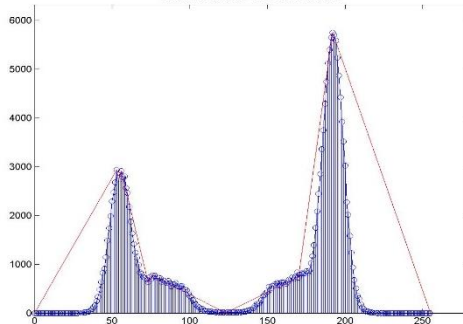


Image after Segmentation



Image after segmentation

Multimodal Histogram

If the image is composed of two type (or more) of light objects on a dark background, three (or more) dominant modes characterize the image histogram

- In such a case the histogram has to be partitioned by multiple thresholds.
- Multilevel thresholding classifies a point (x,y) as belonging to one object class

if $T1 < (x,y) \leq T2$,

to the other object class

if $f(x,y) > T2$

and to the background

if $f(x,y) \leq T1$.

Gray Scale Image - Multimodal

Original Image -Lena

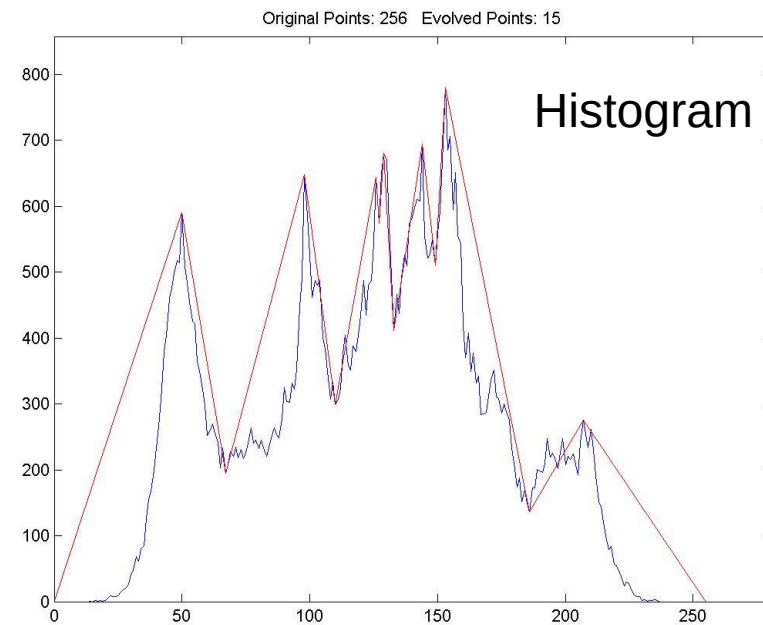


Image after segmentation –
shows the outline of - face, hat,
shadow etc

Problems with this approach

- Noise
- Many holes and discontinuities in the segmentation.
- Changes in the illumination
- we do not use spatial information.
- Crude method- only spectral

Clustering

- There are K clusters $C_1, \dots, C_K$ with means $m_1, \dots, m_K$.
- The **least-squares error** is defined as

$$D = \sum_{k=1}^K \sum_{x_i \in C_k} \| x_i - m_k \|^2 .$$

- Out of all possible partitions into K clusters, choose the one that minimizes D.

Why don't we just do this?

If we could, would we get meaningful objects?

Some Clustering Methods

- K-means Clustering
- Isodata Clustering

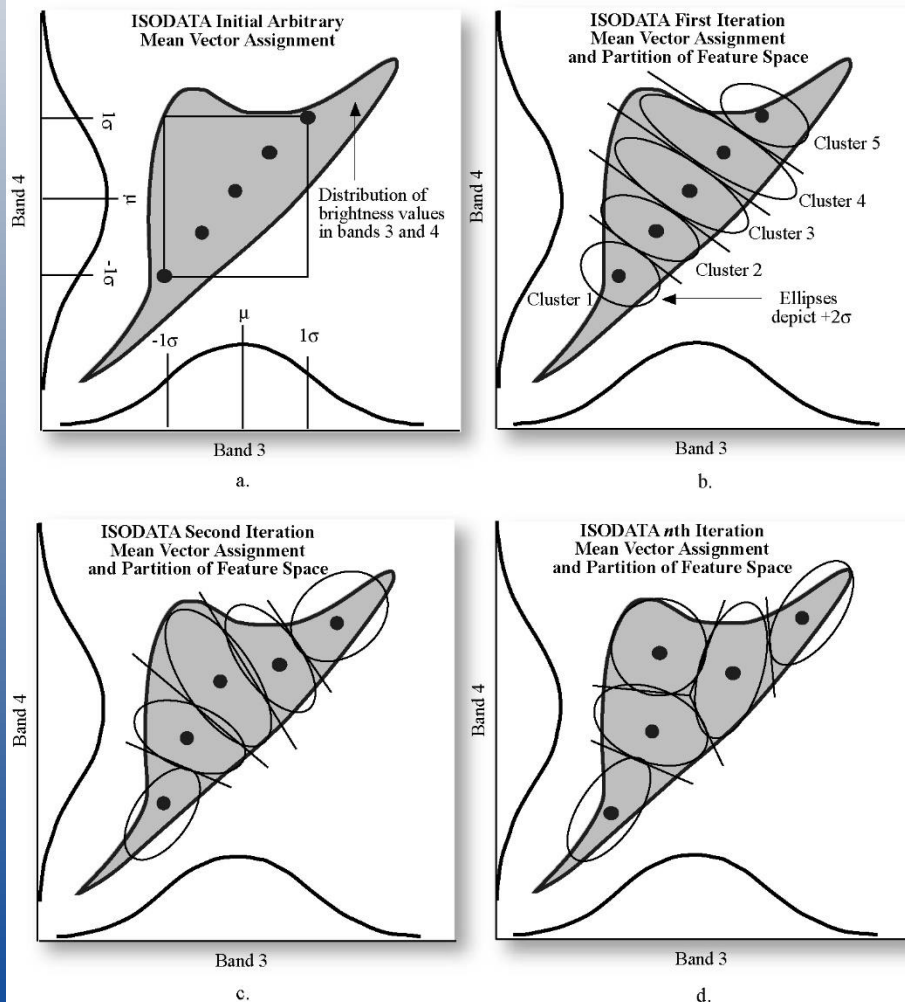
Isodata Clustering

1. Select several cluster means and form clusters.
2. Split any cluster whose variance is too large.
3. Group together clusters that are too small.
4. Recompute means.
5. Repeat till 2 and 3 cannot be applied.

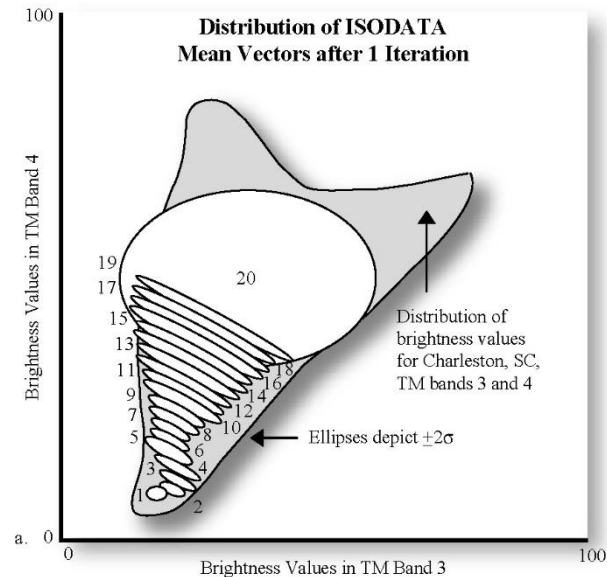
ISODATA algorithm normally requires the analyst to specify-

- **Cmax** : maximum no. of clusters to be identified.
- **T**:maximum % of pixels whose class values are allowed to be unchanged between iterations.
- **M** :maximum no. of times isodata is to classify pixels and recalculate cluster mean vectors.
- Minimum members in a cluster
- Maximum standard deviation for a cluster.
- Split separation value (if the valuse is changed from 0.0, it takes the place of S.D.)
- Minimum distance between cluster means.

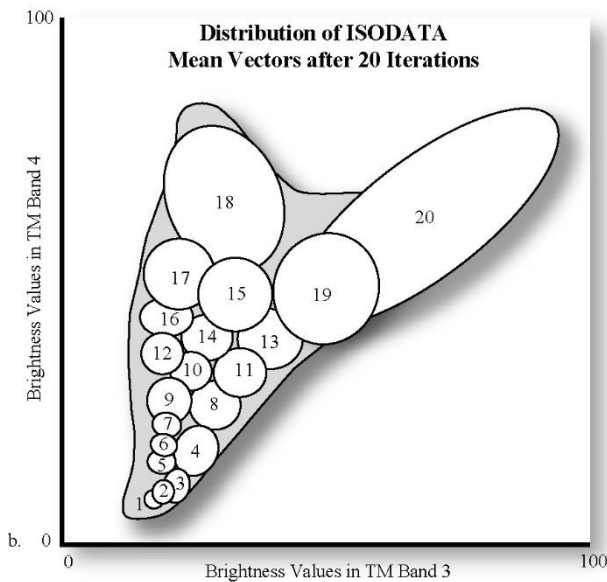
Phase 1: ISODATA Cluster Building using many passes through the dataset.



- ISODATA initial distribution of five hypothetical mean vectors using $\pm 1\sigma$ standard deviations in both bands as beginning and ending points.
- In the first iteration, each candidate pixel is compared to each cluster mean and assigned to the cluster whose mean is closest in Euclidean distance.
- During the second iteration, a new mean is calculated for each cluster based on the actual spectral locations of the pixels assigned to each cluster, instead of the initial arbitrary calculation. This involves analysis of several parameters to merge or split clusters. After the new cluster mean vectors are selected, every pixel in the scene is assigned to one of the new clusters.
- This split-merge-assign process continues until there is little change in class assignment between iterations (the T threshold is reached) or the maximum number of iterations is reached (M).



a) Distribution of 20 *ISODATA* mean vectors after just one iteration



b) Distribution of 20 *ISODATA* mean vectors after 20 iterations. The bulk of the important feature space (the gray background) is partitioned rather well after just 20 iterations.

K-means Clustering

- ❑ Approximation to an NP- hard combinatorial optimization problem.
- ❑ A clustering algorithm.
- ❑ It is unsupervised.
- ❑ K stands for no. of clusters, it is a user input to algorithm.
- ❑ From a set o data or observations, K-means to classify them into k clusters.
- ❑ The algorithm is iterative in nature.
- ❑ K-means converges, but finds local minima of the cost function.
- ❑ Works only for numerical observation.
- ❑ Fine tuning is required when working on image segmentation, because there is no imposed spatial coherency in K-means algorithm.
- ❑ It works as a starting point for sophisticated image segmentation algorithm.

K-Means Example

1. Select an image: 2. Select a processor: 3. Click

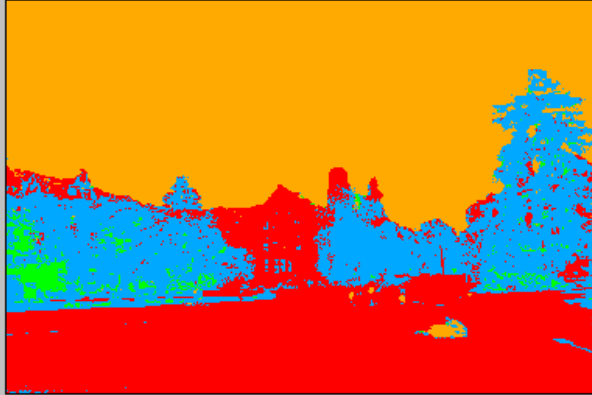


640*480 (607,118): RGB(20,22,1)

Options:

Init Method

Process done !



(228,26): RGB(255,170,0)

1. Select an image: 2. Select a processor: 3. Click

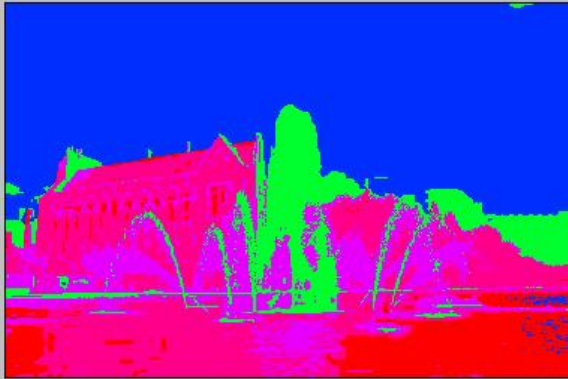


640*480 (636,95): RGB(102,130,151)

Options:

Init Method

Process done !



(590,209): RGB(0,46,255)

Region Growing

Region growing techniques start with one pixel of a potential region and try to **grow** it by adding adjacent pixels till the pixels being compared are too dissimilar.

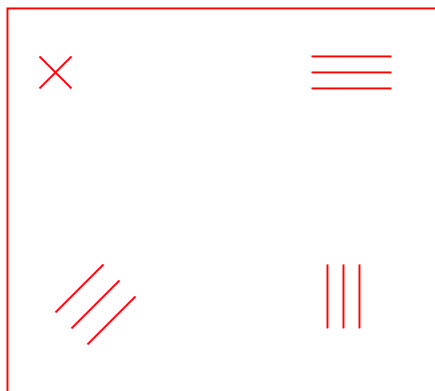
- The first pixel selected can be just the first unlabeled pixel in the image or a set of seed pixels can be chosen from the image.
- Usually a statistical test is used to decide which pixels can be added to a region.

- **Homogeneity Test:**
 - Pixel value $\pm$ threshold
 - Pixel value variation from mean should be within a threshold

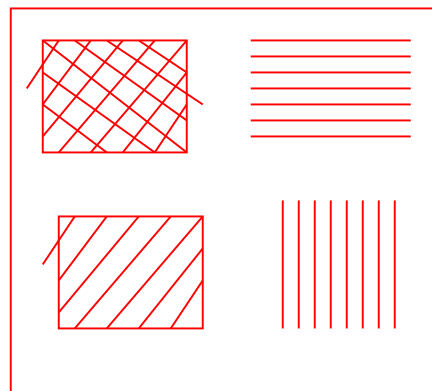
Starts from a uniform seed region.

Region extraction is done by appending adjacent pixels which satisfy a similarity criteria.

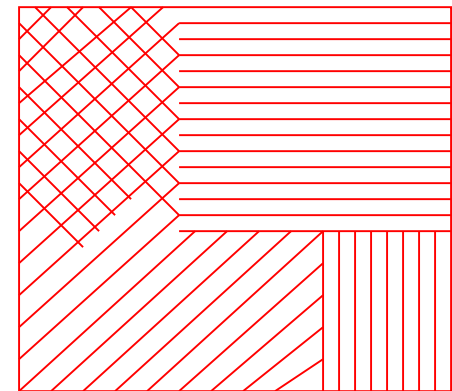
1. Determination of seeds
2. Region growing
3. Region merging



Initial Seeds



First Growth Step



Final Segmentation

Region Growing – contd.

- Homogeneity test: if the pixel intensity is close to the region mean value

$$|I(r,c) - M(i)| \leq T(i)$$

- Threshold T_i varies depending on the region R_n and the intensity of the pixel $I(r,c)$. It can be chosen this way:

$$T(i) = \{ 1 - [s.d(i)/M(i)] \} T$$

Splitting:

- Starts with assumption that the entire image is homogeneous.
- Variance is more than a certain threshold, if not the image split into 4 sub images.
- Results in blocky appearance.

Split-Merge:

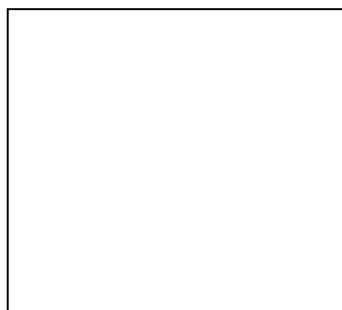
- The opposite approach to region growing-region shrinking (splitting).
- It is a top-down approach and it starts with the assumption that the entire image is homogeneous
- If this is not true , the image is split into four sub images
- This splitting procedure is repeated recursively until we split the image into homogeneous regions

SPLIT AND MERGE

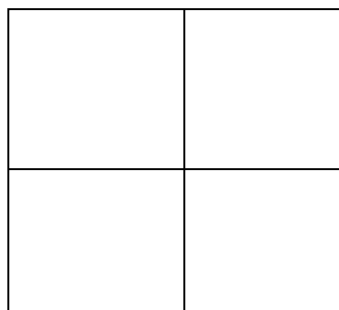
- Regions are examined to see if it is a valid region according to a similarity measure.
- If a region is consistent it is accepted else it is split into halves and quarters and each new region is examined.
- Splitting continues till all regions are consistent.
- Adjacent regions are examined and consistent regions are combined.

Split / Merge

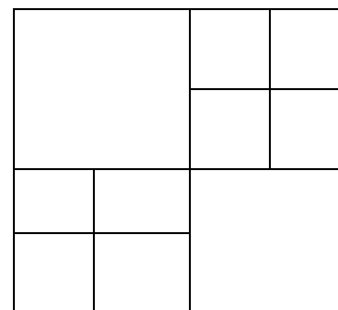
- If the original image is square $N \times N$, having dimensions that are powers of 2 ($N = 2^n$):
- All regions produced by the splitting algorithm are squares having dimensions $M \times M$, where M is a power of 2 as well ($M=2^m, M \leq n$).
- Since the procedure is recursive, it produces an image representation that can be described by a tree whose nodes have four sons each
- Such a tree is called a Quadtree.



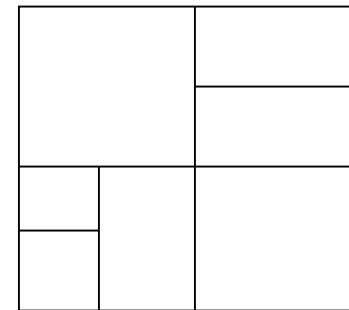
Whole Image



First Split



Second Split



Merged

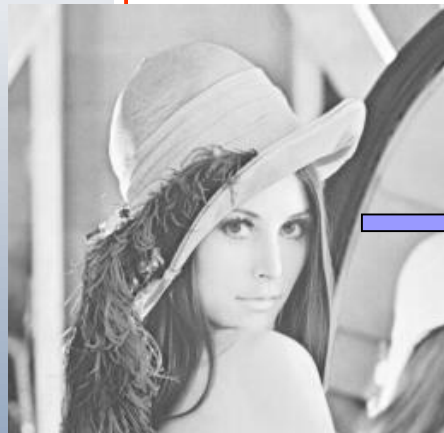
Splitting and Merging

- Characteristic of the split and merge method was the algorithm given by Horwitz and Plavidis.
- The algorithm starts with a cutset of nodes. This consists of five arrays storing the node properties: $\{x_k, y_k, \text{size}, M_k, m_k\}$.
- Merging is done by comparing adjacent groups of four nodes with a common parent. If they obey the criterion they are replaced by their parent node. Similarly splitting is done by breaking non uniform nodes into the four children nodes at the lower level .
- When all the splitting and merging is complete, the final cutset must be grouped into regions.

Split / Merge

- Splitting techniques disadvantage, they create regions that may be adjacent and homogeneous, but not merged.
- Split and Merge method – It is an iterative algorithm that includes both splitting and merging at each iteration.
- The split and merge algorithm produces more compact regions than the pure splitting algorithm

Results



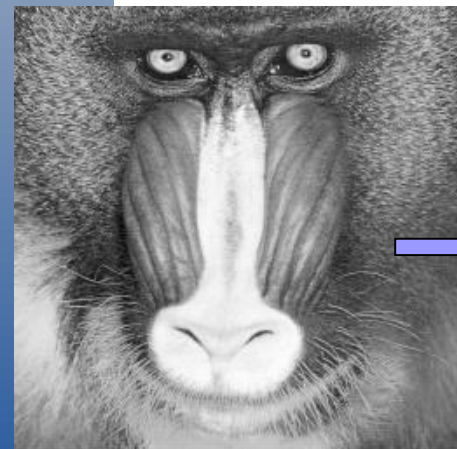
Region Grow



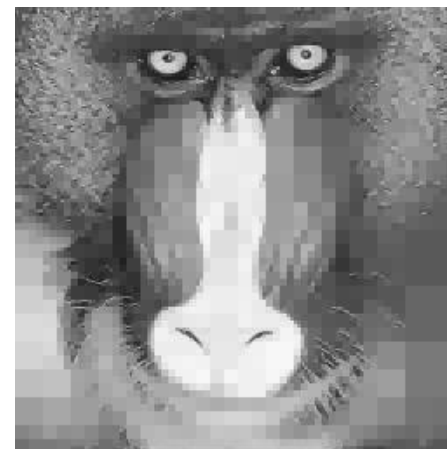
Region Split



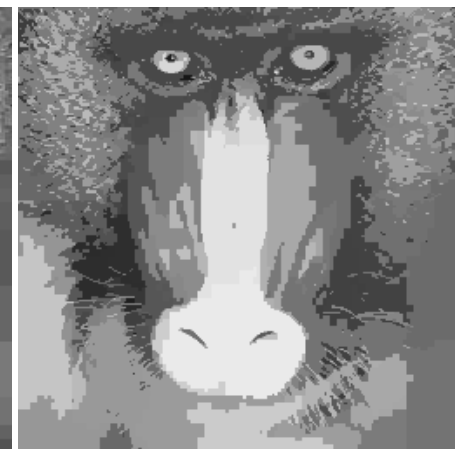
Region Split and Merge



Region Grow



Region Split



Region Split and Merge

Object-oriented Image Analysis and segmentation

- The basic processing units are image objects (segments), not single pixels;
- Uses soft classifiers that are based on fuzzy logic, not hard classifiers.
- Object-oriented segmentation attempts to describe relationships in terms of spectral and spatial information which considers not only the spectral characteristics of a single pixel but also the characteristics of its surrounding pixels.
- Object features (group of pixels) have properties such as color, shape, area, and scale.

Multi-resolution Analysis

- Multi-resolution is a bottom up region-merging technique starting with one-pixel objects.
- In numerous subsequent steps, smaller image objects are merged into bigger ones.
- The merging decision is based on a local homogeneity criterion, describing the similarity between adjacent image objects
- If the smallest growth exceeds the threshold defined by the scale parameter, the process stops.
- Throughout the segmentation procedure, the whole image is segmented and image objects are generated based upon several adjustable criteria of homogeneity in color and shape.

Heterogeneity Criteria:

a. Spectral /colour Hetrogeneity:

Spectral (i.e., color) heterogeneity (h) of an image object is computed as the sum of the standard deviations of spectral values of each layer - σ_k (i.e., band) multiplied by the weights for each layer (w_k):

$$h = \sum_{k=1}^m w_k \cdot \sigma_k$$

The **color criterion** is computed as the weighted mean of all changes in standard deviation for each band k of the m bands of remote sensing dataset. The standard deviation σ_k are weighted by the object sizes n_{ob} (i.e. the number of pixels):

$$h_{color} = \sum_{k=1}^m w_k \left[n_{mg} \cdot \sigma_k^{mg} - (n_{ob1} \cdot \sigma_k^{ob1} + n_{ob2} \cdot \sigma_k^{ob2}) \right]$$

where *mg* means merge.

b. Shape Hetrogeneity:

compactness

$$cpt = \frac{l}{\sqrt{n}}$$

smoothness

$$smooth = \frac{l}{b}$$

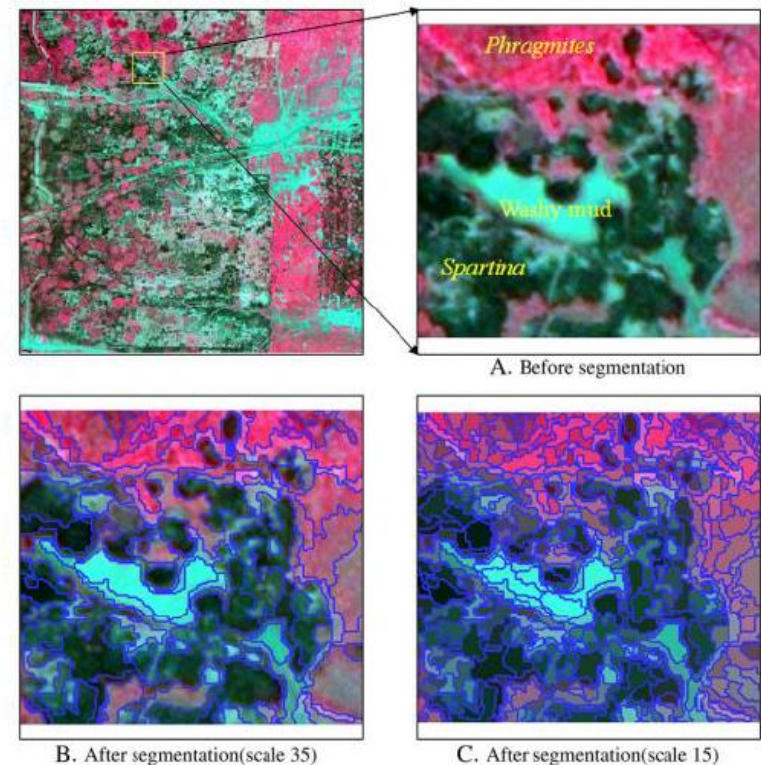
l = border length

n = number of pixels forming length

b = shortest possible border length of a box bounding the object

Multi-resolution Hierarchical Segmentation using Fractal net Evaluation approach (e cognition):

- Represents the idea of hierarchy as fractal net because of self similarity notion of fractals.
- Each coarser level gets the input from finer level.
- If an object is identified at coarse level then it repeats its representation at finer level.
- Process starts with each pixel as object and then merges based on criteria in which merged regions should not exceed a defined heterogeneity threshold.
- Capability to incorporate spectral, texture, spatial, shape, size, prior knowledge and contextual properties of image.

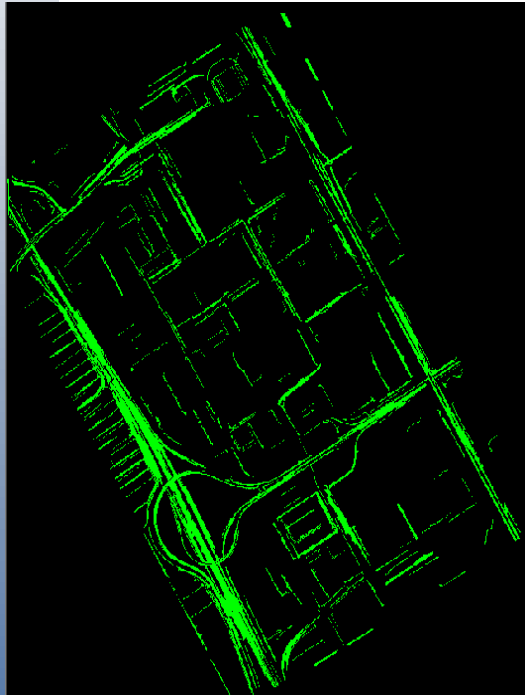


Obeject Oriented Image Analysis Software

- Definiens - ecognition
- Feature Analyst – ENVI
- Imagine Objective
- Few freeware (MultiSpec and SPRING)
- Open source software
 - a. Orfeo ToolBox (OTB) is an open-source project for remote sensing which can be integrated in QGIS. Segmentation algorithms in the toolbox include watershed, meanshift, k-means etc.
 - b. ILWIS is integrated software with raster & vector processing capabilities.
 - c. Ilastik is a user-friendly tool for interactive workflow framework for image classification, segmentation & analysis.
 - d. Parbat – segmentation
 - e. Python libraries

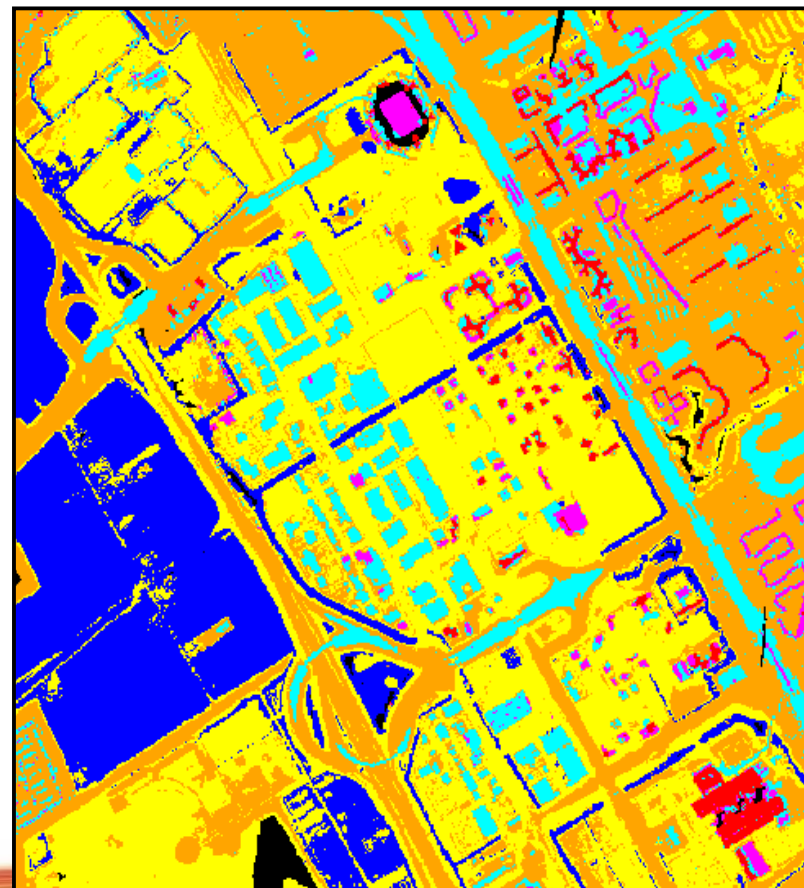
[Recursive Hierarchical Segmentation \(RHSEG\) Project](https://opensource.gsfc.nasa.gov/projects/HSEG/) - GSFC NASA
 (https://opensource.gsfc.nasa.gov/projects/HSEG/)

Automatic Urban Road Extraction Using Airborne Laser Scanning/Altimetry and High Resolution Satellite Data (Poonam S.Tiwari, Ashwini Kumar)



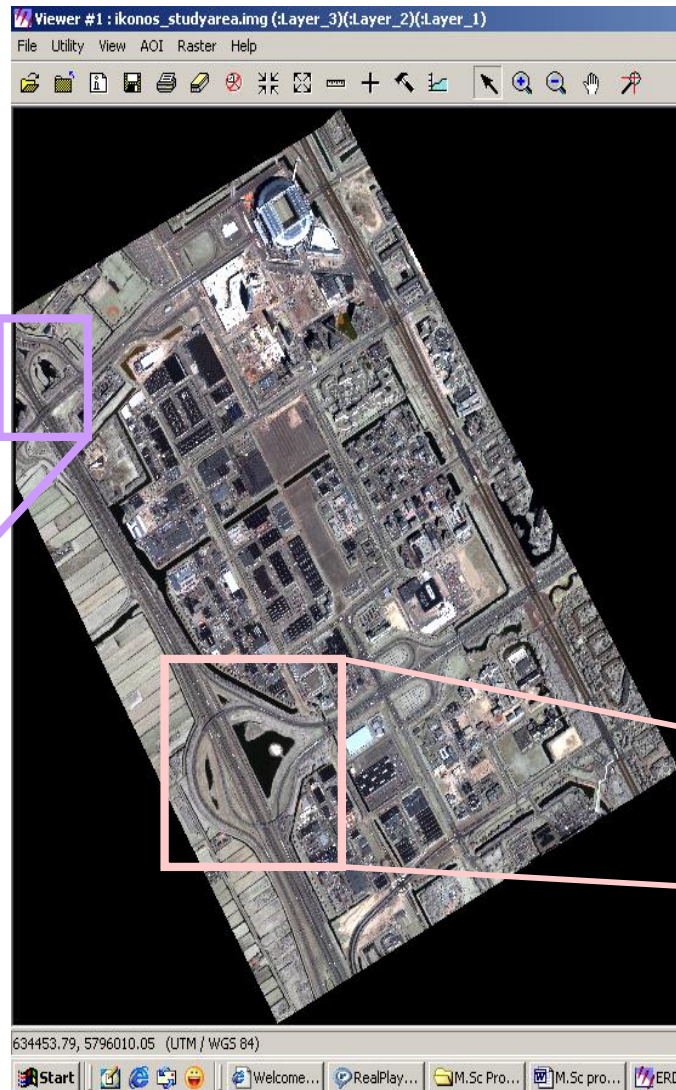
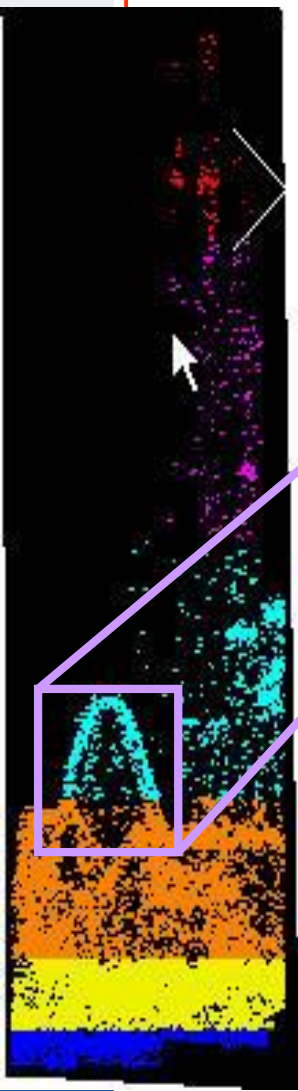
Roads Extracted after Classification

Filtered LiDAR Data



	Very High Buildings
	Medium Rise Buildings
	Flyovers and Low Buildings
	Near Ground
	Ground
	Water
	No Data

Elevation View of Study Area

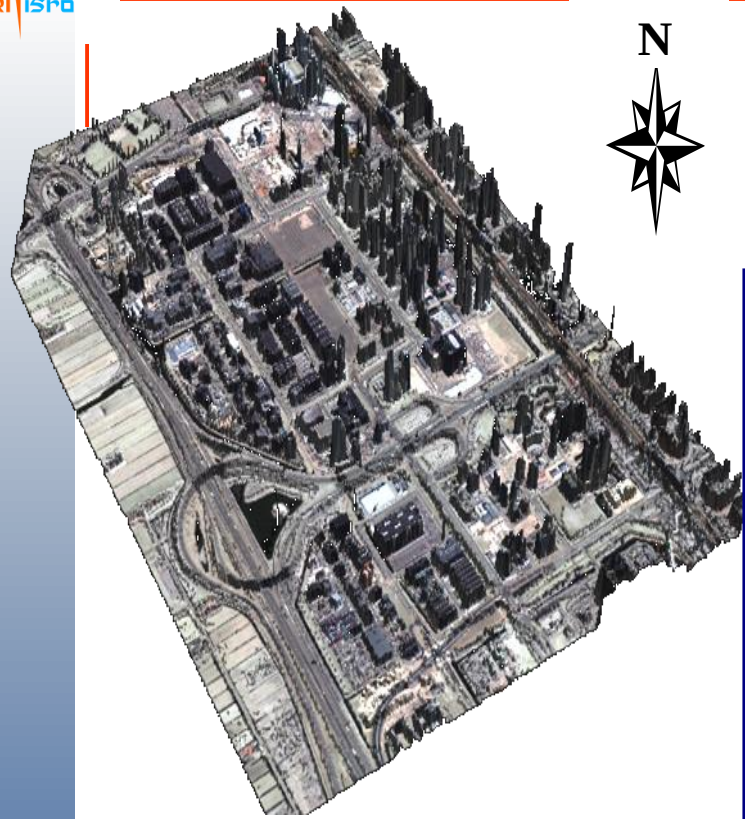


**Very High
Buildings**

**Medium rise
Buildings**

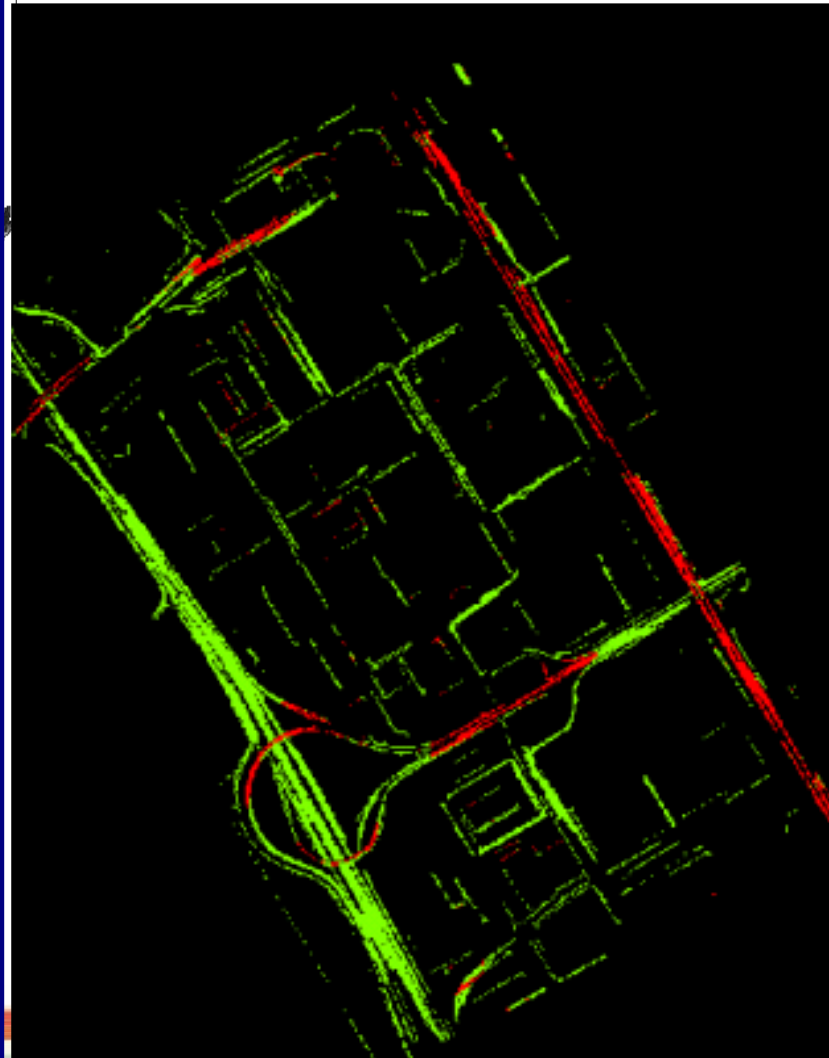
**Flyovers and
Low Buildings**

**Near Ground
Ground
Water**



Perspective View of Study Area

Extracted Road Using LiDAR and IKONOS



Road

Elevate
d Road

HIGH SPATIAL RESOLUTION IMAGERY FOR AUTOMATIC TREE DETECTION

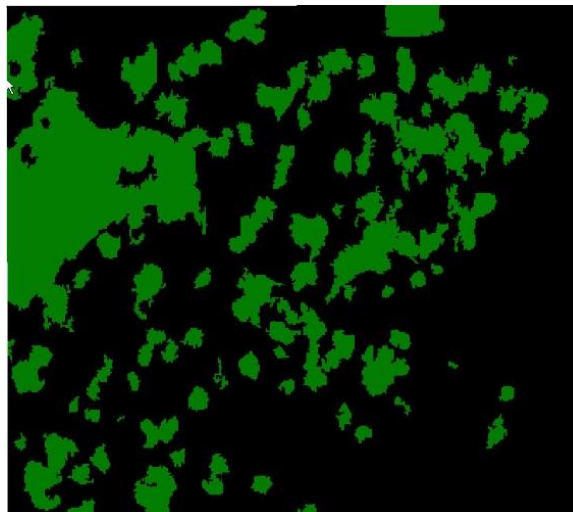
(Hina Pande, Poonam S.Tiwari)

Major Research Questions

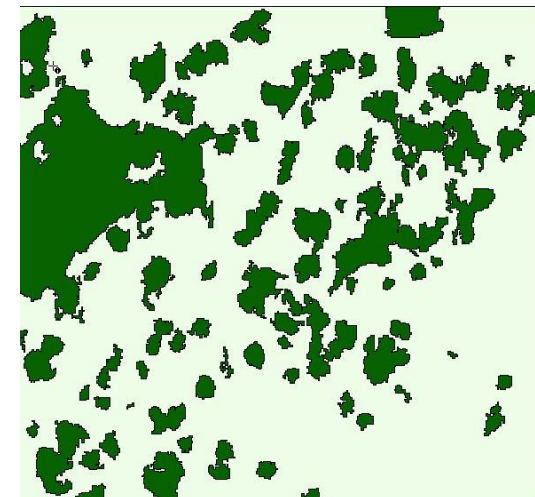
- "How to extract tree canopy boundary from the digital imagery?"
- "How well can this integrated system perform in measuring individual trees semi-automatic or even automatically?"



a.

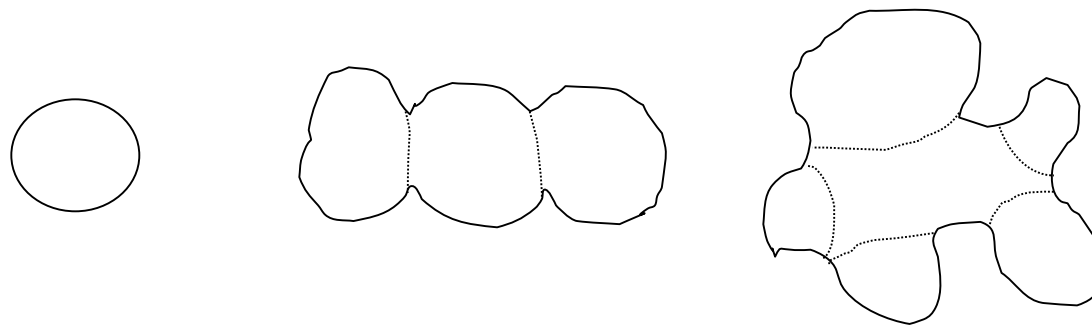


b.



c.

Fig : a: Study Area; b: Extracted tree areas; c: Vectorized trees



a. Isolated tree

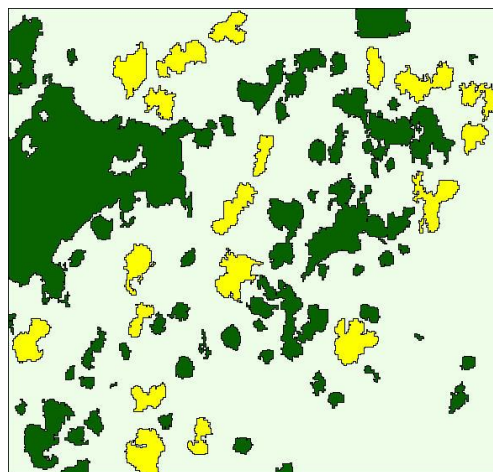
b. Slightly touching trees

c. Tree clumps

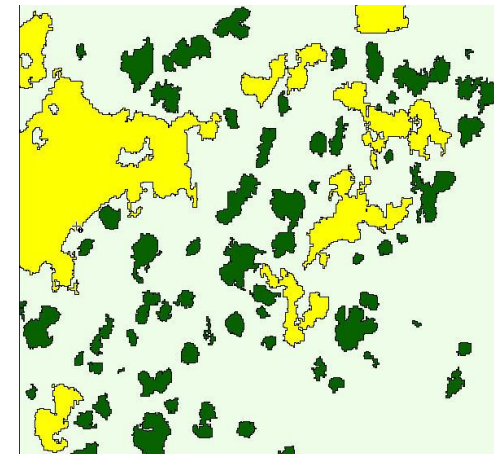
Fig : Three typical cases of trees in the study area



a.



b.



c.

Fig 5: a: Individual trees; b: Slightly touching trees; c: Clump of trees

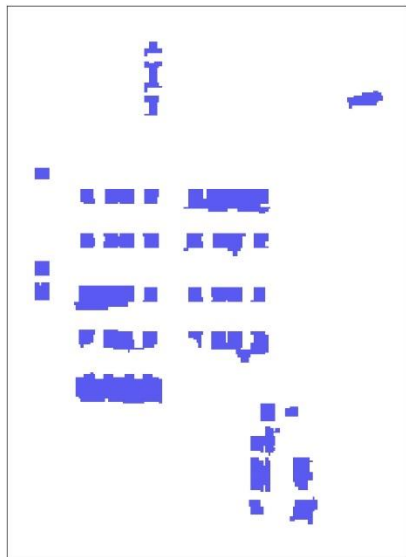
	Visually interpreted	Automatically Extracted
Individual Trees	67	45
Slightly touching Trees	29	22
Tree Clumps	5	7

Numerical Description of Extraction

	AREA MEASURED IN SQ. METERS							
EXTRACTION TYPE	SINGLE TREE	% AREA	SLIGHTLY TOUCHING TREES	% AREA	CLUSTER OF TREES	% AREA	TOTAL AREA	% OF TOTAL TREE AREA TO PLOT AREA
AUTOMATIC	8282	25.79	9401	29.27	14430	44.94	32112	24.63
VISUAL	7184	18.49	11033	28.40	20629	53.10	38846	29.80
DIFFERENCE	-1098		1632		6199		6734	
DIFFERENCE IN PERCENTAGE		7.30		0.87		-8.17		-5.17

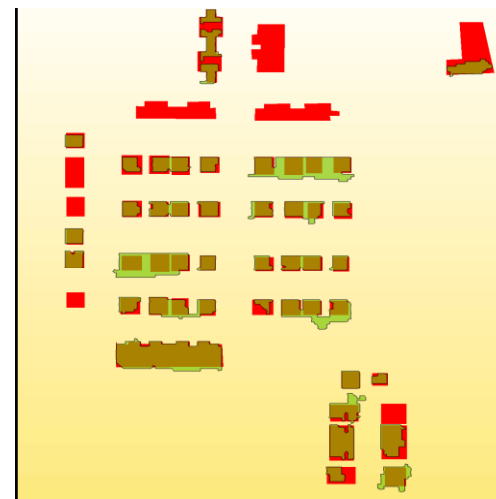
Statistical Evaluation Of Error in Extraction

OBJECT ORIENTED FUZZY RULE BASE CLASSIFICATION (IKONOS)



**AFTER FUZZY RULE BASE
CLASSIFICATION**

Planned Area



LEGEND

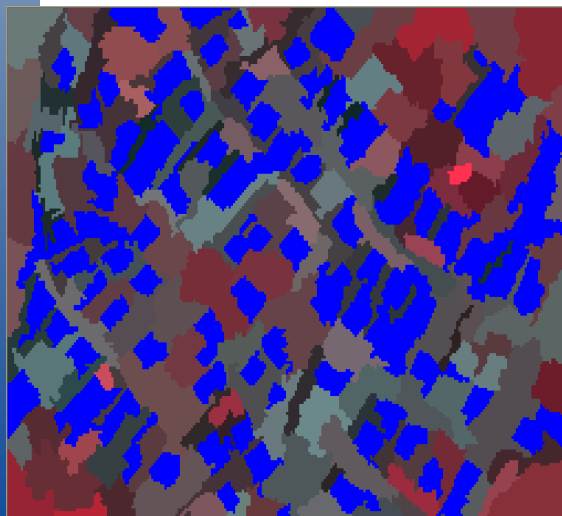


EXTRACTED BUILDINGS



REFERENCE BUILDINGS

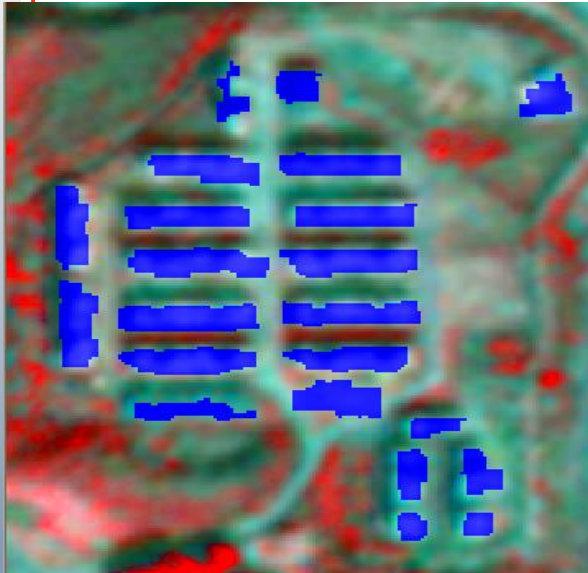
**OVERLAY OF EXTRACTED AND
REFERENCE BUILDINGS**



UnPlanned Area

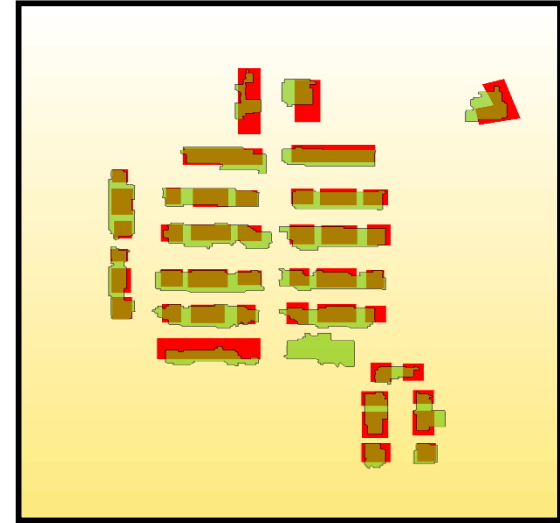


LEVELWISE OBJECT ORIENTED FUZZY RULE BASE CLASSIFICATION(CARTOSAT)



**AFTER FUZZY RULE BASE
CLASSIFICATION**

Planned Area

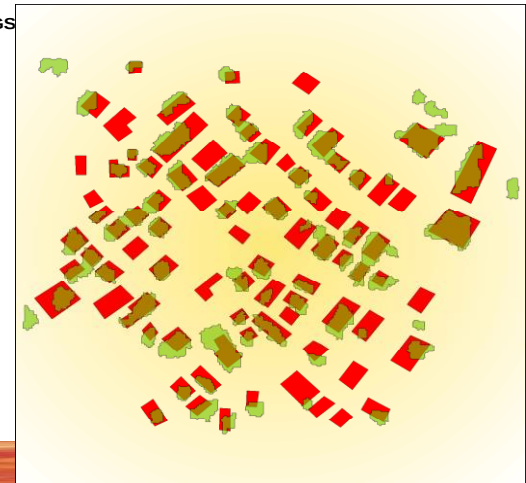
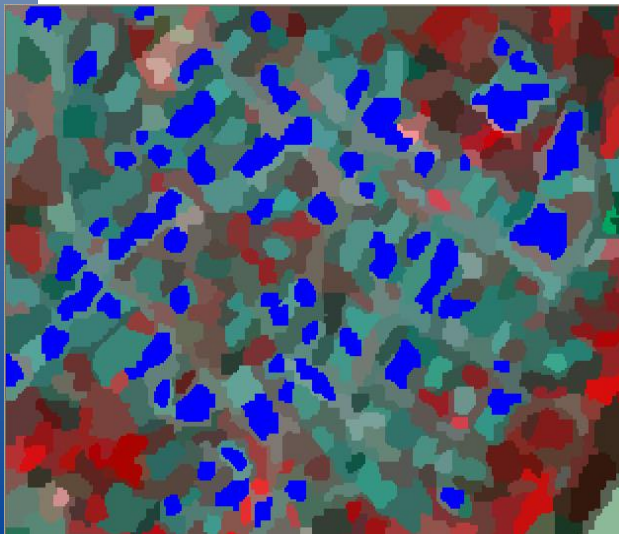


**OVERLAY OF EXTRACTED AND
REFERENCE BUILDINGS**

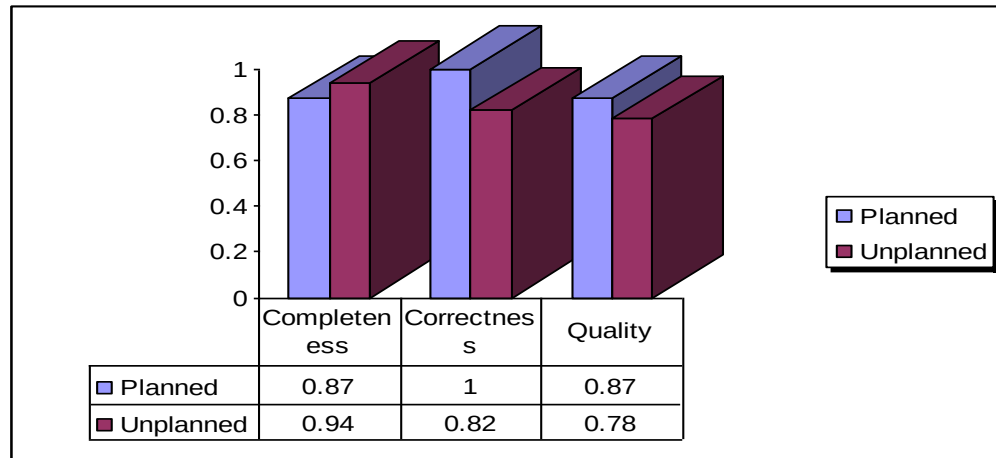
Legend

- EXTRACTED BUILDING
- REFERENCE BUILDINGS

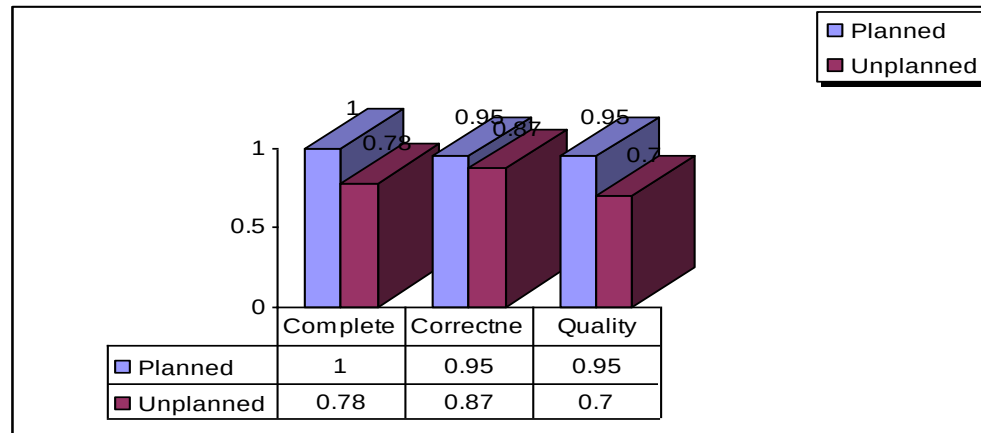
UnPlanned Area



Error Analysis

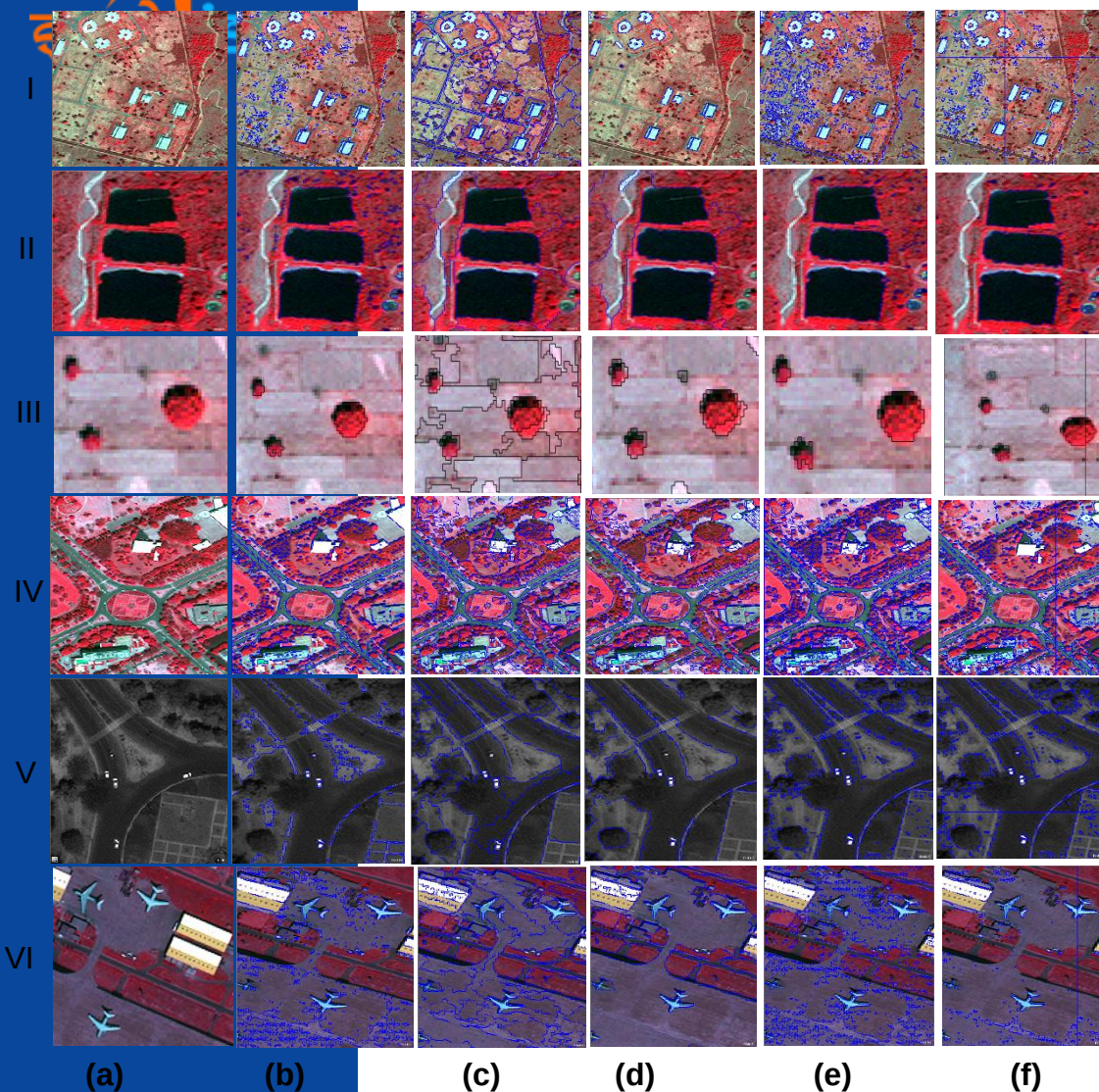


IKONOS



CARTOSAT

Performance Analysis of Segmentation for feature identification



a) actual image (b) Fuzzy segmentation (c) Multi resolution algorithm (d) spectral difference algorithm (e) Multi threshold algorithm (f) contrast filter algorithm

I: Building , II: Water Body, III: Trees, IV: Roads , V: Vehicles, VI: Airplanes

Goodness Measures

1. Distance Index Method
2. Area Based Method
3. Location Based Method

Distance Measure: “closeness” in the defined space to a Ideal segmentation result, in the context of a pre-defined training set.

Area based method:

$$(1) RA_{or} \% = \frac{1}{n} \sum_{i=1}^n \frac{A_0(i)}{A_r} \times 100\%$$

$$(2) RA_{os} \% = \frac{1}{n} \sum_{i=1}^n \frac{A_0(i)}{A_s(i)} \times 100\%$$

$$(3) qr = \frac{1}{n} \sum_{i=1}^n 1 - \frac{A_0(i)}{A_u(i)}$$

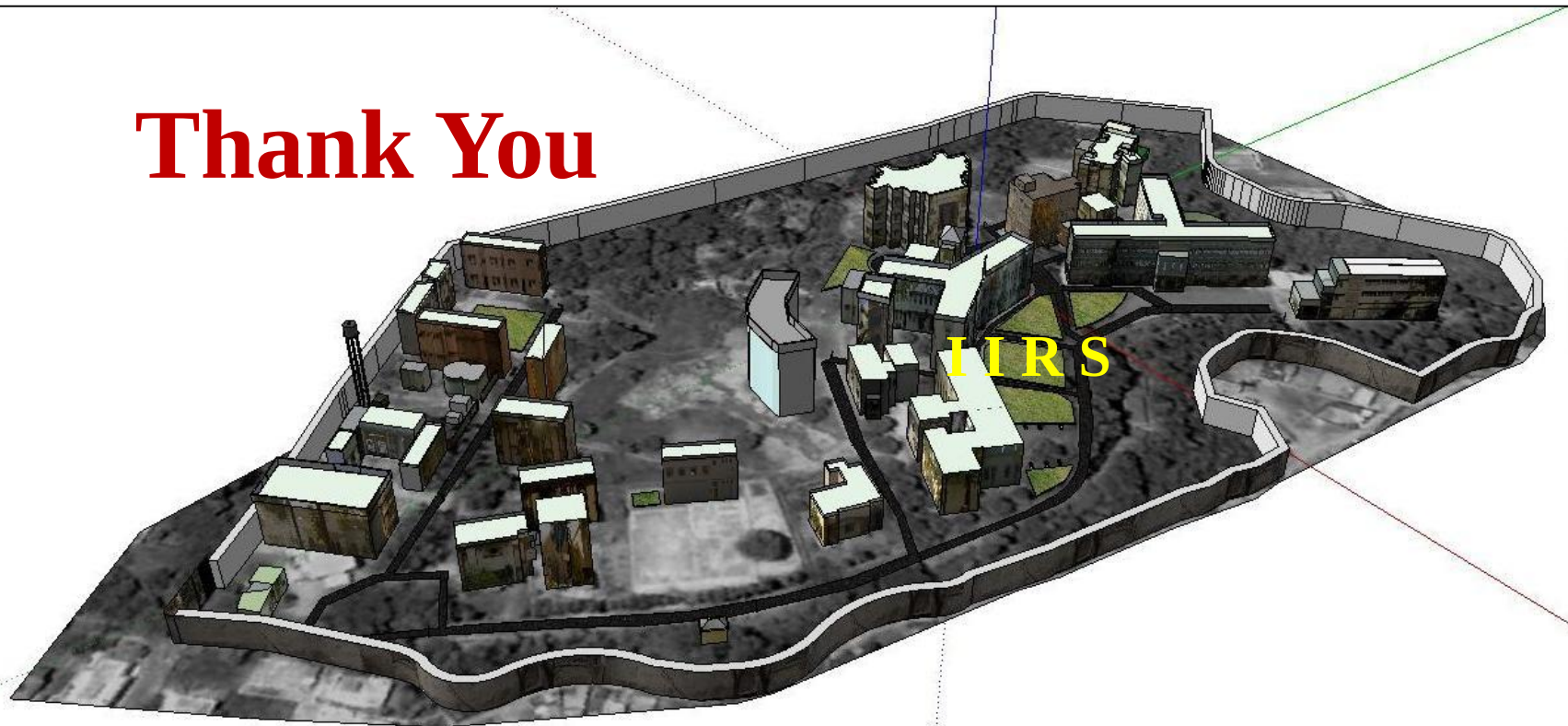
$$(4) SimSize = \frac{1}{n} \sum_{i=1}^n \frac{\min(A_r, A_s(i))}{\max(A_r, A_s(i))}$$

Location based method: Euclidian distance has been computed between the centroid co-ordinates of the reference object and the centroid co-ordinates of the segmented object.

Analysis:

- Features with regular boundaries (man made) give better results with multiresolution segmentation
- Features with irregular boundaries display better results with Fuzzy segmentation.

Thank You



Email: poonam@iirs.gov.in

The material for the presentation has been compiled from various sources- books, tutorials, lecture notes, several resources on the internet